TRAINING REGULATIONS

AUTOMOTIVE DIAGNOSIS (ENGINE) NC III

AUTOMOTIVE AND LAND TRANSPORTATION SECTOR

TECHNICAL EDUCATION AND SKILLS DEVELOPMENT AUTHORITY
TESDA Complex East Service Road, South Luzon Expressway (SLEX), Fort Bonifacio, Taguig City

Technical Education and Skills Development Act of 1994
(Republic Act No. 7796)

Section 22, “Establishment and Administration of the National Trade Skills Standards” of the RA 7796 known as the TESDA Act mandates TESDA to establish national occupational skill standards. The Authority shall develop and implement a certification and accreditation program in which private industry group and trade associations are accredited to conduct approved trade tests, and the local government units to promote such trade testing activities in their respective areas in accordance with the guidelines to be set by the Authority.

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The Training Regulations (TR) serve as basis for the:

1. Competency assessment and certification;
2. Registration and delivery of training programs; and
3. Development of curriculum and assessment instruments.

Each TR has four sections:

Section 1 **Definition of Qualification** - describes the qualification and defines the competencies that comprise the qualification.

Section 2 **Competency Standards** - was revised to include the Required Knowledge and Required Skills per element. These fields explicitly state the required knowledge and skills for competent performance of a unit of competency in an informed and effective manner. These also emphasize the application of knowledge and skills to situations where understanding is converted into a workplace outcome.

Section 3 **Training Arrangements** - contain the information and requirements which serve as bases for training providers in designing and delivering competency-based curriculum for the qualification. The revisions to Section 3 entail identifying the Learning Activities leading to achievement of the identified Learning Outcome.

Section 4 **Assessment and Certification Arrangements** - describe the policies governing assessment and certification procedures for the qualification.

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TRAINING REGULATIONS FOR

AUTOMOTIVE DIAGNOSIS (ENGINE) NC III

SECTION 1 DEFINITION OF QUALIFICATION

The **AUTOMOTIVE DIAGNOSIS (ENGINE) NC III** Qualification consists of competencies that a person must achieve to diagnose and repair: petrol electronic fuel injection system, emission control system, diesel electronic fuel injection systems, forced-induction system, and electronic spark advance system.

This Qualification is packaged from the competency map of the Automotive and Land Transportation Industry (Service sector) as shown in Annex A.

The Units of Competency comprising this qualification include the following

Unit Code	BASIC COMPETENCIES
400311319	Lead workplace communication
400311320	Lead small teams
400311321	Apply critical thinking and problem-solving techniques in the workplace
400311322	Work in a diverse environment
400311323	Propose methods of applying learning and innovation in the organization
400311324	Use information systematically
400311325	Evaluate occupational safety and health work practices
400311326	Evaluate environmental work practices
400311327	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)
Unit Code	**COMMON COMPETENCIES**
ALT723211	Validate vehicle specification
ALT832212	Move and position vehicle
ALT723213	Utilize automotive tools
ALT723214	Perform mensuration and calculation
ALT723215	Utilize workshop facilities and equipment
ALT723216	Prepare servicing parts and consumables
ALT723217	Prepare vehicle for servicing and releasing
Unit Code	**CORE COMPETENCIES**
ALT723381	Diagnose and repair petrol electronic fuel injection system
ALT723399	Diagnose and repair emission control system
ALT7233100	Diagnose and repair diesel electronic fuel injection system
ALT7233101	Diagnose and repair forced-induction system
ALT7233102	Diagnose and repair electronic spark advance system

A person who has achieved this Qualification is competent to be:

- ❑ Automotive Engine Control Specialist

SECTION 2 COMPETENCY STANDARDS

This section gives the details of the contents of the basic, common and core units of competency required in **AUTOMOTIVE DIAGNOSIS (ENGINE) NC III**.

BASIC COMPETENCIES

UNIT OF COMPETENCY : LEAD WORKPLACE COMMUNICATION

UNIT CODE : 400311319

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to lead in the effective dissemination and discussion of ideas, information, and issues in the workplace. This includes preparation of written communication materials.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Communicate information about workplace processes	1.1 Relevant *communication method* is selected based on workplace procedures 1.2 Multiple operations involving several topics/areas are communicated following enterprise requirements 1.3 Questioning is applied to gain extra information 1.4 Relevant sources of information are identified in accordance with workplace/ client requirements 1.5 Information is selected and organized following enterprise procedures 1.6 Verbal and written reporting is	1.1. Organization requirements for written and electronic communication methods 1.2. Effective verbal communication methods 1.3. Business writing 1.4. Workplace etiquette	1.1 Organizing information 1.2 Conveying intended meaning 1.3 Participating in a variety of workplace discussions 1.4 Complying with organization requirements for the use of written and electronic communication methods 1.5 Effective business writing 1.6 Effective clarifying and probing skills 1.7 Effective questioning techniques (clarifying and probing)

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	undertaken when required 1.7 Communication and negotiation skills are applied and maintained in all relevant situations		
2. Lead workplace discussions	2.1 Response to workplace issues are sought following enterprise procedures 2.2 Response to workplace issues are provided immediately 2.3 Constructive contributions are made to *workplace discussions* on such issues as production, quality and safety 2.4 Goals/ objectives and action plans undertaken in the workplace are communicated promptly	2.1 Organization requirements for written and electronic communication methods 2.2 Effective verbal communication methods 2.3 Workplace etiquette	2.1 Organizing information 2.2 Conveying intended meaning 2.3 Participating in variety of workplace discussions 2.4 Complying with organization requirements for the use of written and electronic communication methods 2.5 Effective clarifying and probing skills
3. Identify and communicate issues arising in the workplace	3.1 Issues and problems are identified as they arise 3.2 Information regarding problems and issues are organized coherently to ensure clear and effective communication 3.3 Dialogue is initiated with appropriate personnel 3.4 Communication problems and issues are raised as they arise	3.1 Organization requirements for written and electronic communication methods 3.2 Effective verbal communication methods 3.3 Workplace etiquette 3.4 Communication problems and issues 3.5 Barriers in communication	3.1 Organizing information 3.2 Conveying intended meaning 3.3 Participating in a variety of workplace discussions 3.4 Complying with organization requirements for the use of written and electronic communication methods 3.5 Effective clarifying and probing skills 3.6 Identifying issues

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	3.5 Identify barriers in communication to be addressed appropriately		3.7 Negotiation and communication skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Methods of communication	May include: 1.1. Non-verbal gestures 1.2. Verbal 1.3. Face-to-face 1.4. Two-way radio 1.5. Speaking to groups 1.6. Using telephone 1.7. Written 1.8. Internet
2. Workplace discussions	May include: 2.1. Coordination meetings 2.2. Toolbox discussion 2.3. Peer-to-peer discussion

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Dealt with a range of communication/information at one time 1.2 Demonstrated leadership skills in workplace communication 1.3 Made constructive contributions in workplace issues 1.4 Sought workplace issues effectively 1.5 Responded to workplace issues promptly 1.6 Presented information clearly and effectively written form 1.7 Used appropriate sources of information 1.8 Asked appropriate questions 1.9 Provided accurate information
2. Resource Implications	The following resources should be provided: 2.1 Variety of Information 2.2 Communication tools 2.3 Simulated workplace
3. Methods of Assessment	Competency in this unit may be assessed through: Case problem 3.1. Third-party report 3.2. Portfolio 3.3. Interview 3.4. Demonstration/Role-playing
4. Context for Assessment	4.1. Competency may be assessed in the workplace or in a simulated workplace environment

UNIT OF COMPETENCY : LEAD SMALL TEAMS

UNIT CODE : 400311320

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes to lead small teams including setting, maintaining and monitoring team and individual performance standards.

ELEMENT	**PERFORMANCE CRITERIA** ***Italicized terms*** are elaborated in the Range of Variables	**REQUIRED KNOWLEDGE**	**REQUIRED SKILLS**
1. Provide team leadership	1.1 *Work requirements* are identified and presented to team members based on company policies and procedures 1.2 Reasons for instructions and requirements are communicated to team members based on company policies and procedures 1.3 *Team members' and leaders' concerns* are recognized, discussed and dealt with based on company practices	1.1 Facilitation of Team work 1.2 Company policies and procedures relating to work performance 1.3 Performance standards and expectations 1.4 Monitoring individual's and team's performance vis a vis client's and group's expectations	1.1 Communication skills required for leading teams 1.2 Group facilitation skills 1.3 Negotiating skills 1.4 Setting performance expectation
2. Assign responsibilities	2.1. Responsibilities are allocated having regard to the skills, knowledge and aptitude required to undertake the assigned task based on company policies. 2.2. Duties are allocated having regard to individual preference, domestic and personal considerations, whenever possible	2.1 Work plan and procedures 2.2 Work requirements and targets 2.2 Individual and group expectations and assignments 2.3 Ways to improve group leadership and membership	2.1 Communication skills 2.2 Management skills 2.3 Negotiating skills 2.4 Evaluation skills 2.5 Identifying team member's strengths and rooms for improvement

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Set performance expectations for team members	3.1 Performance expectations are established based on client needs 3.2 Performance expectations are based on individual team members knowledge, skills and aptitude 3.3 Performance expectations are discussed and disseminated to individual team members	3.1 One's roles and responsibilities in the team 3.2 Feedback giving and receiving 3.3 Performance expectation	3.1 Communication skills 3.2 Accurate empathy 3.3 Congruence 3.4 Unconditional positive regard 3.5 Handling of Feedback
4. Supervise team performance	4.1 Performance is monitored based on defined performance criteria and/or assignment instruction 4.2 Team members are provided with *feedback*, positive support and advice on strategies to overcome any deficiencies based on company practices 4.3 *Performance issues* which cannot be rectified or addressed within the team are referred to appropriate personnel according to employer policy 4.4 Team members are kept informed of any changes in the priority allocated to assignments or tasks which might impact on	4.1 Performance Coaching 4.2 Performance management 4.3 Performance Issues	4.1 Communication skills required for leading teams 4.2 Coaching skill

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	client/customer needs and satisfaction 4.5 Team operations are monitored to ensure that employer/client needs and requirements are met 4.6 Follow-up communication is provided on all issues affecting the team 4.7 All relevant documentation is completed in accordance with company procedures		
5. Set performance expectations for team members	5.1 Performance expectations are established based on client needs 5.2 Performance expectations are based on individual team members knowledge, skills and aptitude 5.3 Performance expectations are discussed and disseminated to individual team members	5.1 One's roles and responsibilities in the team 5.2 Feedback giving and receiving 5.3 Performance expectation	5.1 Communication skills 5.2 Accurate empathy 5.3 Congruence 5.4 Unconditional positive regard 5.5 Handling of Feedback
6. Supervise team performance	6.1 Performance is monitored based on defined performance criteria and/or assignment instruction 6.2 Team members are provided with *feedback*, positive support and advice on strategies to	6.1 Performance Coaching 6.2 Performance management 6.3 Performance Issues	6.1 Communication skills required for leading teams 6.2 Coaching skill

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	overcome any deficiencies based on company practices 6.3 *Performance issues* which cannot be rectified or addressed within the team are referred to appropriate personnel according to employer policy 6.4 Team members are kept informed of any changes in the priority allocated to assignments or tasks which might impact on client/customer needs and satisfaction 6.5 Team operations are monitored to ensure that employer/client needs and requirements are met 6.6 Follow-up communication is provided on all issues affecting the team 6.7 All relevant documentation is completed in accordance with company procedures		

RANGE OF VARIABLES

VARIABLE	RANGE
1. Work requirements	Work requirements may include: 1.1. Client Profile 1.2. Assignment instructions
2. Team member's concerns	Team member's concerns may include: 2.1. Roster/shift details
3. Monitor performance	Monitor performance may include: 3.1. Formal process 3.2. Informal process
4. Feedback	Feedback may include: 4.1. Formal process 4.2. Informal process
5. Performance issues	Performance issues may include: 5.1. Work output 5.2. Work quality 5.3. Team participation 5.4. Compliance with workplace protocols 5.5. Safety 5.6. Customer service

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Maintained or improved individuals and/or team performance given a variety of possible scenario 1.2. Assessed and monitored team and individual performance against set criteria 1.3. Represented concerns of a team and individual to next level of management or appropriate specialist and to negotiate on their behalf 1.4. Allocated duties and responsibilities, having regard to individual's knowledge, skills and aptitude and the needs of the tasks to be performed 1.5. Set and communicated performance expectations for a range of tasks and duties within the team and provided feedback to team members
2. Resource Implications	The following resources should be provided: 2.1. Access to relevant workplace or appropriately simulated environment where assessment can take place 2.2. Materials relevant to the proposed activity or task
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Written Examination 3.2. Oral Questioning 3.3. Portfolio
4. Context for Assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : APPLY CRITICAL THINKING AND PROBLEM-SOLVINGTECHNIQUES IN THE WORKPLACE

UNIT CODE : 400311321

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to solve problems in the workplace including the application of problem solving techniques and to determine and resolve the root cause/s of specific problems in the workplace.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Examine specific workplace challenges.	1.1 Variances are examined from normal operating ***parameters*; and** product quality. 1.2 Extent, cause and nature of the specific problem are defined through observation, investigation and ***analytical techniques***. 1.3 ***Problems*** are clearly stated and specified.	1.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations. 1.2 Competence to include the ability to apply and explain, enough for the identification of fundamental causes of specific workplace challenges. 1.3 Relevant equipment and operational processes. 1.4 Enterprise goals, targets and measures. 1.5 Enterprise quality OHS and environmental requirement. 1.6 Enterprise information	1.1 Using range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 1.2 Identifying extent and causes of specific challenges in the workplace.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		systems and data collation 1.7 Industry codes and standards.	
2. Analyze the causes of specific workplace challenges.	2.1 Possible causes of specific problems are identified based on experience and the use of problem solving tools / analytical techniques. 2.2 Possible cause statements are developed based on findings. 2.3 Fundamental causes are identified per results of investigation conducted.	2.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations. 2.2 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendation s. 2.3 Relevant equipment and operational processes. 2.4 Enterprise goals, targets and measures. 2.5 Enterprise quality OSH and environmental requirements. 2.6 Enterprise information systems and data collation. 2.7 Industry codes and standards.	2.1 Using a range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 2.2 Identifying extent and causes of specific challenges in the workplace. 2.3 Providing clear-cut findings on the nature of each identified workplace challenges.
3. Formulate resolutions to	3.1 All possible options are considered for	3.1 Competence to include the ability to apply and	3.1 Using range of analytical techniques (e.g.,

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
specific workplace challenges.	resolution of the problem. 3.2 Strengths and weaknesses of possible options are considered. 3.3 Corrective actions are determined to resolve the problem and possible future causes. 3.4 ***Action plans*** are developed identifying measurable objectives, resource needs and timelines in accordance with safety and operating procedures	explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendation s 3.2Relevant equipment and operational processes 3.3 Enterprise goals, targets and measures 3.4 Enterprise quality OSH and environmental requirement 3.5 Principles of decision making strategies and techniques 3.6 Enterprise information systems and data collation 3.7 Industry codes and standards	planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 3.2 Identifying extent and causes of specific challenges in the workplace. 3.3 Providing clear-cut findings on the nature of each identified workplace challenges. 2.4 Devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges.
4. Implement action plans and communicate results.	4.1 Action plans are implemented and evaluated. 4.2 Results of plan implementation and recommendations are prepared. 4.2 Recommendations are presented to appropriate personnel. 4.3 Recommendations are followed-up, if required.	4.1 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendation s 4.2. Relevant equipment and operational processes	4.1 Using range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 4.2 Identifying extent and causes of specific

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		4.3 Enterprise goals, targets and measures 4.4 Enterprise quality, OSH and environmental requirement 4.5 Principles of decision making strategies and techniques 4.6 Enterprise information systems and data collation 4.7 Industry codes and standards	challenges in the workplace. 4.3 Providing clear-cut findings on the nature of each identified workplace challenges. 4.4 Devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges.

RANGE OF VARIABLES

VARIABLE	RANGE
1. Parameters	Parameters may include: 1.1 Processes 1.2 Procedures 1.3 Systems
2. Analytical techniques	Analytical techniques may include: 2.1. Brainstorming 2.2. Intuitions/Logic 2.3. Cause and effect diagrams 2.4. Pareto analysis 2.5. SWOT analysis 2.6. Gant chart, Pert CPM and graphs 2.7. Scattergrams
3. Problem	Problem may include: 3.1. Routine, non – routine and complex workplace and quality problems 3.2. Equipment selection, availability and failure 3.3. Teamwork and work allocation problem 3.4. Safety and emergency situations and incidents 3.5. Risk assessment and management
4. Action plans	Action plans may include: 4.1. Priority requirements 4.2. Measurable objectives 4.3. Resource requirements 4.4. Timelines 4.5. Co-ordination and feedback requirements 4.6. Safety requirements 4.7. Risk assessment 4.8. Environmental requirements

EVIDENCE GUIDE

1. Critical aspects of Competency	**Assessment requires evidence that the candidate:** 1.1. Examined specific workplace challenges. 1.2. Analyzed the causes of specific workplace challenges. 1.3. Formulated resolutions to specific workplace challenges. 1.4. Implemented action plans and communicated results on specific workplace challenges.
2. Resource Implications	2.1. Assessment will require access to an operating plant over an extended period of time, or a suitable method of gathering evidence of operating ability over a range of situations. A bank of scenarios / case studies / what ifs will be required as well as bank of questions which will be used to probe the reason behind the observable action.
3. Methods of Assessment	**Competency in this unit may be assessed through:** 3.1. Observation 3.2. Case Formulation 3.3. Life Narrative Inquiry 3.4. Standardized test The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components. These assessment activities should include a range of problems, including new, unusual and improbable situations that may have happened.
4. Context for Assessment	In all workplace, it may be appropriate to assess this unit concurrently with relevant teamwork or operation units.

UNIT OF COMPETENCY : WORK IN A DIVERSE ENVIRONMENT

UNIT CODE : 400311322

UNIT DESCRIPTOR : This unit covers the outcomes required to work effectively in a workplace characterized by diversity in terms of religions, beliefs, races, ethnicities and other differences.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Develop an individual's cultural awareness and sensitivity	1.1 Individual differences with clients, customers and fellow workers are recognized and respected in accordance with enterprise policies and core values. 1.2 Differences are responded to in a sensitive and considerate manner 1.3 ***Diversity*** is accommodated using appropriate verbal and non-verbal communication.	1.1 Understanding cultural diversity in the workplace 1.2 Norms of behavior for interacting and dialogue with specific groups (e. g., Muslims and other non-Christians, non-Catholics, tribes/ethnic groups, foreigners) 1.3 Different methods of verbal and non-verbal communication in a multicultural setting	1.1 Applying cross-cultural communication skills (i.e. different business customs, beliefs, communication strategies) 1.2 Showing affective skills – establishing rapport and empathy, understanding, etc. 1.3 Demonstrating openness and flexibility in communication 1.4 Recognizing diverse groups in the workplace and community as defined by divergent culture, religion, traditions and practices
2. Work effectively in an environment that acknowledges and values cultural diversity	2.1 Knowledge, skills and experiences of others are recognized and documented in relation to team objectives.	2.1 Value of diversity in the economy and society in terms of Workforce development 2.2 Importance of inclusiveness in a	2.1 Demonstrating cross-cultural communication skills and active listening 2.2 Recognizing diverse groups in the workplace

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.2 Fellow workers are encouraged to utilize and share their specific qualities, skills or backgrounds with other team members and clients to enhance work outcomes. 2.3 Relations with customers and clients are maintained to show that diversity is valued by the business.	diverse environment 2.3 Shared vision and understanding of and commitment to team, departmental, and organizational goals and objectives 2.4 Strategies for customer service excellence	and community as defined by divergent culture, religion, traditions and practices 2.3 Demonstrating collaboration skills 2.4 Exhibiting customer service excellence
3. Identify common issues in a multicultural and diverse environment	3.1 ***Diversity-related conflicts*** within the workplace are effectively addressed and resolved. 3.2 Discriminatory behaviors towards customers/stakeholders are minimized and addressed accordingly. 3.3 Change management policies are in place within the organization.	3.1 Value, and leverage of cultural diversity 3.2 Inclusivity and conflict resolution 3.3 Workplace harassment 3.4 Change management and ways to overcome resistance to change 3.5 Advanced strategies for customer service excellence	3.1 Addressing diversity-related conflicts in the workplace 3.2 Eliminating discriminatory behavior towards customers and co-workers 3.3 Utilizing change management policies in the workplace

RANGE OF VARIABLES

VARIABLE	RANGE
1. Diversity	This refers to diversity in both the workplace and the community and may include divergence in : 1.1 Religion 1.2 Ethnicity, race or nationality 1.3 Culture 1.4 Gender, age or personality 1.5 Educational background
2. Diversity-related conflicts	May include conflicts that result from: 2.1 Discriminatory behaviors 2.2 Differences of cultural practices 2.3 Differences of belief and value systems 2.4 Gender-based violence 2.5 Workplace bullying 2.6 Corporate jealousy 2.7 Language barriers 2.8 Individuals being differently-abled persons 2.9 Ageism (negative attitude and behavior towards old people)

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Adjusted language and behavior as required by interactions with diversity 1.2 Identified and respected individual differences in colleagues, clients and customers 1.3 Applied relevant regulations, standards and codes of practice
2. Resource Implications	The following resources should be provided: 2.1 Access to workplace and resources 2.2 Manuals and policies on Workplace Diversity
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration or simulation with oral questioning 3.2 Group discussions and interactive activities 3.3 Case studies/problems involving workplace diversity issues 3.4 Third-party report 3.5 Written examination 3.6 Role Plays
4. Context for Assessment	Competency assessment may occur in workplace or any appropriately simulated environment

UNIT OF COMPETENCY : **PROPOSE METHODS OF APPLYING LEARNING AND INNOVATION IN THE ORGANIZATION**

UNIT CODE : **400311323**

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to assess general obstacles in the application of learning and innovation in the organization and to propose practical methods of such in addressing organizational challenges.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Assess work procedures, processes and systems in terms of innovative practices.	1.1. ***Reasons*** for innovation are incorporated to work procedures. 1.2. ***Models of innovation*** are researched. 1.3. ***Gaps or barriers*** to innovation in one's work area are analyzed. 1.4. Staff who can support and foster innovation in the work procedure are identified.	1.1 Seven habits of highly effective people. 1.2 Character strengths that foster innovation and learning (Christopher Peterson and Martin Seligman, 2004) 1.3 Five minds of the future concepts (Gardner, 2007). 1.4 Adaptation concepts in neuroscience (Merzenich, 2013). 1.5 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	1.1 Demonstrating collaboration and networking skills. 1.2 Applying basic research and evaluation skills 1.3 Generating insights on how to improve organizational procedures, processes and systems through innovation.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Generate practical action plans for improving work procedures, processes.	2.1 Ideas for innovative work procedure to foster innovation using individual and group techniques are conceptualized 2.2 Range of ideas with other team members and colleagues are evaluated and discussed 2.3 Work procedures and processes subject to change are selected based on ***workplace requirements*** (feasible and innovative). 2.4 Practical action plans are proposed to facilitate simple changes in the work procedures, processes and systems. 2.5 ***Critical inquiry*** is applied and used to facilitate discourse on adjustments in the simple work procedures, processes and systems.	2.1 Seven habits of highly effective people. 2.2 Character strengths that foster innovation and learning (Christopher Peterson and Martin Seligman, 2004) 2.3 Five minds of the future concepts (Gardner, 2007). 2.4 Adaptation concepts in neuroscience (Merzenich, 2013). 2.5 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	2.1 Assessing readiness for change on simple work procedures, processes and systems. 2.2 Generating insights on how to improve organizational procedures, processes and systems through innovation. 2.3 Facilitating action plans on how to apply innovative procedures in the organization.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Evaluate the effectiveness of the proposed action plans.	3.1 Work structure is analyzed to identify the impact of the new work procedures 3.2 Co-workers/key personnel is consulted to know who will be involved with or affected by the work procedure 3.3 Work instruction operational plan of the new work procedure is developed and evaluated. 3.4 Feedback and suggestion are recorded. 3.5 Operational plan is updated. 3.6 Results and impact on the developed work instructions are reviewed 3.7 Results of the new work procedure are evaluated 3.8 Adjustments are recommended based on results gathered	3.1 Five minds of the future concepts (Gardner, 2007). 3.2 Adaptation concepts in neuroscience (Merzenich, 2013). 3.3 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	3.1 Generating insights on how to improve organizational procedures, processes and systems through innovation. 3.2 Facilitating action plans on how to apply innovative procedures in the organization. 3.3 Communicating results of the evaluation of the proposed and implemented changes in the workplace procedures and systems. 3.4 Developing action plans for continuous improvement on the basic systems, processes and procedures in the organization.

RANGE OF VARIABLES

VARIABLE	RANGE
1. Reasons	May include: 1.1 Strengths and weaknesses of the current systems, processes and procedures. 1.2 Opportunities and threats of the current systems, processes and procedures.
2. Models of innovation	May include: 2.1 Seven habits of highly effective people. 2.2 Five minds of the future concepts (Gardner, 2007). 2.3 Neuroplasticity and adaptation strategies.
3. Gaps or barriers	May include: 3.1 Machine 3.2 Manpower 3.3 Methods 3.4 Money
4. Critical Inquiry	May include: 4.1 Preparation. 4.2 Discussion. 4.3 Clarification of goals. 4.4 Negotiate towards a Win-Win outcome. 4.5 Agreement. 4.6 Implementation of a course of action. 4.7 Effective verbal communication. See our pages: Verbal Communication and Effective Speaking. 4.8 Listening. 4.9 Reducing misunderstandings is a key part of effective negotiation. 4.10 Rapport Building. 4.11 Problem Solving. 4.12 Decision Making. 4.13 Assertiveness. 4.14 Dealing with Difficult Situations.

EVIDENCE GUIDE

1. Critical aspects of Competency	**Assessment requires evidence that the candidate:** 1.1 Established the reasons why innovative systems are required 1.2 Established the goals of a new innovative system 1.3 Analyzed current organizational systems to identify gaps and barriers to innovation. 1.4 Assessed work procedures, processes and systems in terms of innovative practices. 1.5 Generate practical action plans for improving work procedures, and processes. 1.6 Reviewed the trial innovative work system and adjusted reflect evaluation feedback, knowledge management systems and future planning. 1.7 Evaluated the effectiveness of the proposed action plans.
2. Resource Implications	The following resources should be provided: 2.1 Pens, papers and writing implements. 2.2 Cartolina. 2.3 Manila papers.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Psychological and behavioral Interviews. 3.2 Performance Evaluation. 3.3 Life Narrative Inquiry. 3.4 Review of portfolios of evidence and third-party workplace reports of on-the-job performance. 3.5 Sensitivity analysis. 3.6 Organizational analysis. 3.7 Standardized assessment of character strengths and virtues applied.
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions.

UNIT OF COMPETENCY : USE INFORMATION SYSTEMATICALLY

UNIT CODE : 400311324

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to use technical information systems, apply information technology (IT) systems and edit, format and check information.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Use technical information	1.1. ***Information*** are *c*ollated and organized into a suitable form for reference and use 1.2. Stored information are classified so that it can be quickly identified and retrieved when needed 1.3. Guidance are advised and offered to people who need to find and use information	1.1. Application in collating information 1.2. Procedures for inputting, maintaining and archiving information 1.3. Guidance to people who need to find and use information 1.4. Organize information 1.5. classify stored information for identification and retrieval 1.6. Operate the technical information system by using agreed procedures	1.1. Collating information 1.2. Operating appropriate and valid procedures for inputting, maintaining and archiving information 1.3. Advising and offering guidance to people who need to find and use information 1.4. Organizing information into a suitable form for reference and use 1.5. Classifying stored information for identification and retrieval 1.6. Operating the technical information system by using agreed procedures

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Apply information technology (IT)	2.1. ***Technical information*** system is operated using agreed procedures 2.2. Appropriate and valid procedures are operated for inputting, maintaining and archiving information 2.3. ***Software*** required are utilized to execute the project activities 2.4. Information and data obtained are handled, edited, formatted and checked from a range of internal and external ***sources*** 2.5. Information are extracted, entered, and processed to produce the outputs required by ***customers*** 2.6. Own skills and understanding are shared to help others 2.7. Specified ***security measures*** are implemented to protect the confidentiality and integrity of project data held in IT systems	2.1. Attributes and limitations of available software tools 2.2. Procedures and work instructions for the use of IT 2.3. Operational requirements for IT systems 2.4. Sources and flow paths of data 2.5. Security systems and measures that can be used 2.6. Extract data and format reports 2.7. Methods of entering and processing information 2.8. WWW enabled applications	2.1. Identifying attributes and limitations of available software tools 2.2. Using procedures and work instructions for the use of IT 2.3. Describing operational requirements for IT systems 2.4. Identifying sources and flow paths of data 2.5. Determining security systems and measures that can be used 2.6. Extracting data and format reports 2.7. Describing methods of entering and processing information 2.8. Using WWW applications

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Edit, format and check information	3.1 Basic editing techniques are used 3.2 Accuracy of documents are checked 3.3 Editing and formatting tools and techniques are used for more complex documents 3.4 Proof reading techniques is used to check that documents look professional	3.1 Basic file-handling techniques 3.2 Techniques in checking documents 3.3 Techniques in editing and formatting 3.4 Proof reading techniques	3.1 Using basic file-handling techniques is used for the software 3.2 Using different techniques in checking documents 3.3 Applying editing and formatting techniques 3.4 Applying proof reading techniques

RANGE OF VARIABLES

VARIABLE	RANGE
1. Information	May include: 1.1. Property 1.2. Organizational 1.3. Technical reference
2. Technical information	May include: 2.1. paper based 2.2. electronic
3. Software	May include: 3.1. spreadsheets 3.2. databases 3.3. word processing 3.4. presentation
4. Sources	May include: 4.1. other IT systems 4.2. manually created 4.3. within own organization 4.4. outside own organization 4.5. geographically remote
5. Customers	May include: 5.1. colleagues 5.2. company and project management 5.3. clients
6. Security measures	May include: 6.1. access rights to input; 6.2. passwords; 6.3. access rights to outputs; 6.4. data consistency and back-up; 6.5. recovery plans

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Used technical information systems and information technology 1.2. Applied information technology (IT) systems 1.3. Edited, formatted and checked information
2. Resource Implications	The following resources MUST be provided: 2.1. Computers 2.2. Software and IT system
3. Methods of Assessment	Competency in this unit MUST be assessed through: 3.1. Direct Observation 3.2. Oral interview and written test
4. Context for Assessment	4.1. Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : EVALUATE OCCUPATIONAL SAFETY AND HEALTH WORK PRACTICES

UNIT CODE : 400311325

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to interpret Occupational Safety and Health practices, set OSH work targets, and evaluate effectiveness of Occupational Safety and Health work instructions.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Interpret Occupational Safety and Health practices	1.1 ***OSH work practices issues*** are identified relevant to work requirements 1.2 OSH work standards and procedures are determined based on applicability to nature of work 1.3 Gaps in work practices are identified related to relevant OSH work standards	1.1. OSH work practices issues 1.2. OSH work standards 1.3. General OSH principles and legislations 1.4. Company/ workplace policies/ guidelines 1.5. Standards and safety requirements of work process and procedures	1.1. Communication skills 1.2. Interpersonal skills 1.3. Critical thinking skills 1.4. Observation skills
2. Set OSH work targets	2.1 Relevant work information are gathered necessary to determine OSH work targets 2.2 ***OSH Indicators*** based on gathered information are agreed upon to measure effectiveness of workplace OSH policies and procedures 2.3 Agreed OSH indicators are endorsed for approval from appropriate personnel 2.4 ***OSH work instructions*** are	2.1. OSH work targets 2.2. OSH Indicators 2.3. OSH work instructions 2.4. Safety and health requirements of tasks 2.5. Workplace guidelines on providing feedback on OSH and security concerns 2.6. OSH regulations Hazard control procedures 2.7. OSH trainings relevant to work	2.1. Communication skills 2.2. Collaborating skills 2.3. Critical thinking skills 2.4. Observation skills

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	received in accordance with workplace policies and procedures*		
3. Evaluate effectiveness of Occupational Safety and Health work instructions	3.1 OSH Practices are observed based on workplace standards 3.2 Observed OSH practices are measured against approved ***OSH metrics*** 3.3 Findings regarding effectiveness are assessed and gaps identified are implemented based on OSH work standards	3.1. OSH Practices 3.2. OSH metrics 3.3. OSH Evaluation Techniques 3.4. OSH work standards	3.1. Critical thinking skills 3.2. Evaluating skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. OSH Work Practices Issues	May include: 1.1 Workers' experience/observance on presence of work hazards 1.2 Unsafe/unhealthy administrative arrangements (prolonged work hours, no break-time, constant overtime, scheduling of tasks) 1.3 Reasons for compliance/non-compliance to use of PPEs or other OSH procedures/policies/ guidelines
2. OSH Indicators	May include: 2.1 Increased of incidents of accidents, injuries 2.2 Increased occurrence of sickness or health complaints/symptoms 2.3 Common complaints of workers' related to OSH 2.4 High absenteeism for work-related reasons
3. OSH Work Instructions	May include: 3.1 Preventive and control measures, and targets 3.2 Eliminate the hazard (i.e., get rid of the dangerous machine 3.3 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 3.4 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 3.5 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OSH-related topics, issue warning signages, rotation/shifting work schedule) 3.6 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 3.7 Use personal protective equipment 3.8 Safety, Health and Work Environment Evaluation 3.9 Periodic and/or special medical examinations of workers
4. OSH metrics	May include: 4.1 Statistics on incidence of accidence and injuries 4.2 Morbidity (Type and Number of Sickness) 4.3 Mortality (Cause and Number of Deaths) 4.4 Accident Rate

EVIDENCE GUIDE

1. Critical aspects of Competency	**Assessment requires evidence that the candidate:** 1.1. Identify OSH work practices issues relevant to work requirements 1.2. Identify gaps in work practices related to relevant OSH work standards 1.3. Agree upon OSH Indicators based on gathered information to measure effectiveness of workplace OSH policies and procedures 1.4. Receive OSH work instructions in accordance with workplace policies and procedures 1.5. Compare Observed OSH practices with against approved OSH work instructions 1.6. Assess findings regarding effectiveness based on OSH work standards
2. Resource Implications	**The following resources should be provided:** 2.1 Facilities, materials, tools and equipment necessary for the activity
3. Methods of Assessment	**Competency in this unit may be assessed through:** 3.1 Observation/Demonstration with oral questioning 3.2 Third party report 3.3 Written exam
4. Context for Assessment	4.1 Competency may be assessed in the work place or in a simulated work place setting

UNIT OF COMPETENCY : EVALUATE ENVIRONMENTAL WORK PRACTICES

UNIT CODE : 400311326

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitude to interpret environmental Issues, establish targets to evaluate environmental practices and evaluate effectiveness of environmental practices.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Interpret environmental practices, policies and procedures	1.1 ***Environmental work practices*** issues are identified relevant to work requirements 1.2 Environmental Standards and Procedures nature of work are determined based on Applicability to nature of work ~~1.3~~ Gaps in work practices related to Environmental Standards and Procedures are identified	1.1 Environmental Issues 1.2 Environmental Work Procedures 1.3 Environmental Laws 1.4 Environmental Hazardous and Non-Hazardous Materials 1.5 Environmental required license, registration or certification	1.1. Analyzing Environmental Issues and Concerns 1.2. Critical thinking 1.3. Problem Solving 1.4. Observation Skills
2. Establish targets to evaluate environmental practices	2.1. Relevant information are gathered necessary to determine environmental work targets 2.2. ***Environmental Indicators*** based on gathered information are set to measure environmental work targets 2.3. Indicators are verified with appropriate personnel	2.1. Environmental Indicators 2.2. Relevant Environment Personnel or expert 2.3. Relevant Environmental Trainings and Seminars	2.1. Investigative Skills 2.2. Critical thinking 2.3. Problem Solving 2.4. Observation Skills

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Evaluate effectiveness of environmental practices	3.1. Work environmental practices are recorded based on workplace standards 3.2. Recorded work environmental practices are compared against planned indicators 3.3. Findings regarding effectiveness are assessed and gaps identified are implemented based on environment work standards and procedures 3.4. Results of environmental assessment are conveyed to appropriate personnel	1.1. Environmental Practices 1.2. Environmental Standards and Procedures	3.1 Documentation and Record Keeping Skills 3.2 Critical thinking 3.3 Problem Solving 3.4 Observation Skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Environmental Practices Issues	May include: 1.1 Water Quality 1.2 National and Local Government Issues 1.3 Safety 1.4 Endangered Species 1.5 Noise 1.6 Air Quality 1.7 Historic 1.8 Waste 1.9 Cultural
2. Environmental Indicators	May include: 2.1 Noise level 2.2 Lighting (Lumens) 2.3 Air Quality - Toxicity 2.4 Thermal Comfort 2.5 Vibration 2.6 Radiation 2.7 Quantity of the Resources 2.8 Volume

EVIDENCE GUIDE

1. Critical aspects of Competency	**Assessment requires evidence that the candidate:** 1.1. Identified environmental issues relevant to work requirements 1.2. Identified gaps in work practices related to Environmental Standards and Procedures 1.3. Gathered relevant information necessary to determine environmental work targets 1.4. Set environmental indicators based on gathered information to measure environmental work targets 1.5. Recorded work environmental practices are recorded based on workplace standards 1.6. Conveyed results of environmental assessment to appropriate personnel
2. Resource Implications	**The following resources should be provided:** 2.1 Workplace/Assessment location 2.2 Legislation, policies, procedures, protocols and local ordinances relating to environmental protection 2.3 Case studies/scenarios relating to environmental protection
3. Methods of Assessment	**Competency in this unit may be assessed through:** 3.1 Written/ Oral Examination 3.2 Interview/Third Party Reports 3.3 Portfolio (citations/awards from GOs and NGOs, certificate of training – local and abroad) 3.4 Simulations and role-plays
4. Context for Assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA center.

UNIT OF COMPETENCY : **FACILITATE ENTREPRENUERIAL SKILLS FOR MICRO-SMALL-MEDIUM ENTERPRISES (MSMEs)**

UNIT CODE : **400311327**

UNIT DESCRIPTOR : This unit covers the outcomes required to build, operate and grow a micro/small-scale enterprise.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Develop and maintain micro-small-medium enterprise (MSMEs) skills in the organization	1.1 Appropriate ***business strategies*** are determined and set for the enterprise based on current and emerging business environment. 1.2 ***Business operations*** are monitored and controlled following established procedures. 1.3 Quality assurance measures are implemented consistently. 1.4 Good relations are maintained with staff/ workers. 1.5 Policies and procedures on occupational safety and health and environmental concerns are constantly observed.	1.1 Business models and strategies 1.2 Types and categories of businesses 1.3 Business operation 1.4 Basic Bookkeeping 1.5 Business internal controls 1.6 Basic quality control and assurance concepts 1.7 Government and regulatory processes	1.1 Basic bookkeeping/ accounting skills 1.2 Communication skills 1.3 Building relations with customer and employees 1.4 Building competitive advantage of the enterprise
2. Establish and Maintain client-base/market	2.1 Good customer relations are maintained 2.2 New customers and markets are identified, explored and reached out to. 2.3 Promotions/ Incentives are	2.1 Public relations concepts 2.2 Basic product promotion strategies 2.3 Basic market and feasibility studies 2.4 Basic business ethics	2.1 Building customer relations 2.2 Individual marketing skills 2.3 Using basic advertising (posters/ tarpaulins, flyers,

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	offered to loyal customers 2.4 Additional products and services are evaluated and tried where feasible. 2.5 ***Promotional/advertising initiatives*** are carried out where necessary and feasible.		social media, etc.)
3. Apply budgeting and financial management skills	3.1 Enterprise is built up and sustained through judicious control of cash flows. 3.2 Profitability of enterprise is ensured though appropriate ***internal controls***. 3.3 Unnecessary or lower-priority expenses and purchases are avoided.	3.1 Cash flow management 3.1 Basic financial management 3.2 Basic financial accounting 3.3 Business internal controls	3.1 Setting business priorities and strategies 3.2 Interpreting basic financial statements 3.3 Preparing business plans

RANGE OF VARIABLES

VARIABLE	RANGE
1. Business strategies	May include: 1.1. Developing/Maintaining niche market 1.2. Use of organic/healthy ingredients 1.3. Environment-friendly and sustainable practices 1.4. Offering both affordable and high-quality products and services 1.5. Promotion and marketing strategies (e. g., on-line marketing)
2. Business operations	May include: 2.1 Purchasing 2.2 Accounting/Administrative work 2.3 Production/Operations/Sales
3. Internal controls	May include: 3.1 Accounting systems 3.2 Financial statements/reports 3.3 Cash management
4. Promotional/Advertising initiatives	May include: 4.1 Use of tarpaulins, brochures, and/or flyers 4.2 Sales, discounts and easy payment terms 4.3 Use of social media/Internet 4.4 "Service with a smile" 4.5 Extra attention to regular customers

EVIDENCE GUIDE

1. Critical aspects of competency	**Assessment requires evidence that the candidate :** 1.1 Demonstrated basic entrepreneurial skills 1.2 Demonstrated ability to conceptualize and plan a micro/small enterprise 1.3 Demonstrated ability to manage/operate a micro/small-scale business
2. Resource Implications	The following resources should be provided: 2.1 Simulated or actual workplace 2.2 Tools, materials and supplies needed to demonstrate the required tasks 2.3 References and manuals
3. Methods of Assessment	**Competency in this unit may be assessed through :** 3.1 Written examination 3.2 Demonstration/observation with oral questioning 3.3 Portfolio assessment with interview 3.4 Case problems
4. Context of Assessment	1.1 Competency may be assessed in workplace or in a simulated workplace setting 1.2 Assessment shall be observed while tasks are being undertaken whether individually or in-group

COMMON COMPETENCIES

UNIT OF COMPETENCY : VALIDATE VEHICLE SPECIFICATION

UNIT CODE : ALT723211

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitude to check body type of the vehicle, check vehicle engine type, check vehicle specifications and complete validation of vehicle specification.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Check body type of the vehicle	1.1 ***Kind of vehicle*** is determined according to job order. 1.2 ***Vehicle dimensions*** is determined according to manual. 1.3 ***Vehicle weight*** is determined according to the manual. 1.4 ***Body shape*** is determined according to the manual. 1.6 ***Power train*** is determined according to the manual. 1.7 Safety practices are applied following OSHS	● Kind of vehicle - Aerodynamics - Vehicle Dynamics - Body shapes - Power train - Major dimensions ● Vehicle specifications - Vehicle performance - Weight & Measurements ● Automotive history ● Documentation/ Accomplishing checklist ● Resources information - Bulletin - Shop manual ● OSHS ● PPEs Attitude: ● Patience ● Attention to details	● Identifying kind of vehicle, dimensions, weight, body shape, and power train ● Accomplishing checklist ● Estimating visually dimensions and masses ● Utilizing resource information ● Wearing PPEs ● Applying safety practices
2. Check vehicle engine type	2.1 ***Engine type*** is identified according to industry standards. 2.2 Engine ***fuel/energy system*** is identified according to manual. 2.3 ***Engine components*** are	● Principles of internal combustions ● Principles of Electricity and motors ● History of engines ● Hybrid technology ● Resources information - Bulletin - Shop manual	● Identifying engine type, parts & components ● Identifying fuel systems or energy systems ● Utilizing resource information

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	identified following manual.		
3. Check vehicle specifications	3.1 VIN plate is inspected for specification of vehicle according to manual. 3.2 Vehicle specification is verified according to ***vehicle reference materials***. 3.3 Vehicle modifications and conversions are checked following the manual. 3.3 Vehicle conversions are inspected following the manual.	● Fundamentals of Automotive engineering: - Understanding of power & torque - Gear Ratios - Vehicle Regulations - Knowledge of vehicle performance - Knowledge in Vehicle manufacturing process - Knowledge of vehicle use - Automotive history ● Knowledge in specifications ● Reading of brochure, owner's manuals ● Reading of Resources information - Bulletin - Shop manual	● Reading vehicle reference materials ● Conducting vehicle inspection for modification and conversion ● Comparing actual vehicle and specification sheets ● Utilizing resource information
4. Complete validation of vehicle specification	4.1 Vehicle ownership is verified using repair order and vehicle reference materials. 4.2 ***Dealers check sheet*** is accomplished following industry standards. 4.3 ***Dealers check sheet*** is submitted to immediate superior following industry standards.	● Reporting to immediate superior ● Documentation/ Accomplishing checklist Attitude: ● Accuracy	● Verifying vehicle ownership ● Accomplishing dealers check sheet ● Reporting skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Kind of Vehicle	May include: 1.1 Motorized 1.2 Not Motorized 1.3 On-Road 1.4 Off-Road 1.5 Passenger 1.6 Commercial 1.7 Utility 1.8 Manned 1.9 Unmanned 1.10 Remote control 1.11 Automated/Self Driving 1.12 Guided
2. Vehicle Dimensions	May include: 2.1 Overall length 2.2 Overall width 2.3 Overall height 2.4 Wheelbase 2.5 Tread 2.6 Minimum running ground clearance 2.7 Room Length 2.8 Room Width 2.9 Room Height 2.10 Overhang front 2.11 Overhang rear 2.12 Angle of approach 2.13 Angle of departure
3. Vehicle Weight	May include: 3.1 Gross weight 3.2 Curb weight 3.3 Tare weight 3.4 Net weight
4. Body Shape	May include: 4.1 Sedan 4.2 Coupe 4.3 Hardtop 4.4 Convertible 4.5 Multipurpose vehicle (MPV) 4.6 Sports utility vehicle (SUV) 4.7 Truck 4.8 Tractor Head 4.9 Trailer 4.10 Special Utility Truck 4.11 Bus 4.12 Mini Bus 4.13 Articulated bus 4.14 Asian Utility Vehicle (AUV)

VARIABLE	RANGE
5. Power Train	May include: 5.1 Front Wheel Drive 5.2 Rear Wheel Drive 5.3 4x2 5.4 4x4 5.5 Limited Slip Differential (LSD) 5.6 Manual Transmission 5.7 Automatic Transmission 5.8 Continuously Variable Transmission
6. Engine Type	May include: 6.1 Internal Combustion Engine 6.2 Electric Motor
7. Fuel/Energy System	May include: 7.1 Diesel Fuel 7.2 Gasoline Fuel 7.3 Compressed Natural Gas (CNG) 7.4 Liquefied Petroleum Gas (LPG) 7.5 Methanol 7.6 Hydrogen 7.7 Biodiesel 7.8 Solar Cell 7.9 Fuel Cell
8. Engine Components	May include: 8.1 Intake System 8.2 Electrical System 8.3 Cooling System 8.4 Exhaust System 8.5 Valve Train System 8.6 Cylinder Head 8.7 Engine Block 8.8 Lubricating System
9. Vehicle reference materials	May include: 9.1 Warranty booklet 9.2 Brochure of the vehicle 9.3 Vehicle registration
10. Dealers check sheet	May include: 10.1 Vehicle mileage 10.2 Owner's information 10.3 Damage

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Checked body type of the vehicle 1.2 Checked vehicle engine type 1.3 Checked vehicle specifications 1.4 Completed validation of vehicle specification
2. Resource Implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate vehicle or model equivalent 2.3 Materials relevant to the activity 2.4 Resource information, references, and manual
3. Method of Assessment	Competency in this unit may be assessed through: 3.1 Direct Observation 3.2 Interview 3.3 Third Party Report 3.4 Written exam 3.5 Demonstration with Oral questioning
4. Context of Assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : MOVE AND POSITION VEHICLE

UNIT CODE : ALT832212

UNIT DESCRIPTOR : This unit involves the skills and knowledge and attitudes required to move and position vehicle safely including systematic and efficient control of all vehicle functions.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare vehicle for operation	1.1 Vehicle ***multi point inspection*** is conducted according to industry practice. 1.2 ***Cockpit Drill*** is performed according to industry practice. 1.3 Vehicle is start-up following owner's manual. 1.4 Parking brake is engaged according to industry practice.	• Revolutions per minute during idle • Manual, automatic and CVT Transmission • Vehicle parts, components and functions • Inspection procedures • Owner's manual • Safety procedures	• Performing Cockpit Drill • Conducting Vehicle Multi point inspection • Starting the engine • Using owner's manual
2. Position vehicle	2.1 ***Workshop hazards*** are identified and avoided as per standard operating procedures. 2.2 Vehicle is moved according to Occupational Health and Safety Standards. 2.3 Workshop rules and regulations are recognized according to standard procedures.	• Revolutions per minute in running condition • Kilometer per hour • Estimation/ timing • Manual, automatic and CVT Transmission • Diesel, Gasoline and EV engines • Vehicle parts, components and functions • Defensive driving • Owner's Manual • Safety procedures	• Skills in positioning vehicle • Vehicle positioning estimation skill • Identifying workshop signs and markings
3. Park and stop the vehicle	3.1 Vehicle is positioned according to ***parking rules and regulations***.	• Vehicle parts, components and functions • Inspection procedures • Owner's Manual	• Vehicle positioning estimation skills • Identifying parking signs and markings

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	3.2 Parking brake is engaged according to industry practice. 3.3 ***Electrical devices*** are turned off based on manufacturer's specification. 3.4 Vehicle is shut-off following owner's manual	● Procedure in shutting-off vehicle ● Safety procedures ● Parking rules and regulations	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Multi point inspection	May include: 1.1 Check for any obstruction 1.2 Check external condition 1.3 Check internal condition 1.3.1 Manual transmission 1.3.2 Automatic transmission 1.4 Check vehicle drivability
2. Cockpit Drill	May include: 2.1 Car mirror adjustments 2.3 Steering the car 2.4 How to change gears 2.5 Use of parking brake 2.6 Doors, Seat, Steering, Seat belt and Mirrors 2.7 Foot controls 2.8 Hand controls 2.9 Auxiliary controls (indicators)
3. Workshop hazards	May include: 3.1 Workshop tools and materials 3.2 Workshop equipment 3.3 Other vehicles 3.4 Other people 3.5 Oil spills 3.6 Loose parts
4. Parking rules and regulation	May include: 4.1 Parallel parking 4.2 Horizontal parking 4.3 Park facing the wall
5. Electrical devices	May include: 5.1 Lights 5.2 Air conditioning 5.3 Wiper 5.4 Radio

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared vehicle for operation 1.2 Positioned the vehicle 1.3 Parked and stopped the vehicle 1.4 Used owner's manual
2. Resource implication	The following resources MUST be provided: 2.1 Workshop range/area 2.2 Service working bay 2.3 Appropriate vehicle for moving and positioning 2.4 Owner's manual
3. Method of assessment	Competency MUST be assessed through: 3.1 Demonstration with oral questioning 3.2 Written exam 3.3 Interview 3.3 Direct observation
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : UTILIZE AUTOMOTIVE TOOLS

UNIT CODE : ALT723213

UNIT DESCRIPTOR : This unit covers the knowledge and skills in selecting and using automotive power tools, hand tools and tool keeping.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare automotive tools	1.1 ***Automotive tools*** are identified according to their classification and specification. 1.2 ***Automotive tools and attachments*** are selected according to job requirements. 1.3 ***Automotive tools and attachments*** are inspected for ***defects and damages*** according to manufacturers and work place procedures. 1.4 Safety practices are applied following OSHS.	● Understanding power to size ratio ● Leverage ● Types of power tools and hand tools ● Uses of automotive power tools and hand tools ● Defects and damages of automotive tools and attachments ● Handling of tools ● Interpretation of contents of users manuals ● Safety procedures ● Wearing of PPE	● Identifying defects or damages of tools before use ● Knowledgeable in proper handling of tools ● Identifying tools required for the job ● Inspecting the area were power tools will be use.
2. Use automotive tools	2.1 ***Attachments*** are mounted to ***automotive tools*** according to job requirements 2.2 Power tools are connected to ***power sources*** according to operation's manual 2.3 Power tools are operated according to operation's manual 2.4 ***Hand tools*** are utilized according to operation's manual	● Use of automotive tools ● Application of Torque and pressure ● Unit conversion of torque ● English and metric system ● Types of hand tools ● Types of power tools ● Fundamentals of automotive hand tools and power tools	● Analytical skills ● Technical literacy ● Mounting attachments to automotive tools ● Connecting power tools to power sources ● Operating power tools ● Utilizing hand tools ● Wearing PPEs ● Applying safety practices ● Following manuals

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.5 ***PPEs*** are worn in accordance to OSHS.	● Interpretation of contents of users manuals ● OSHS ● Resources information - Bulletin - Shop manual	
3. Maintain automotive tools	3.1 ***Automotive tools and attachments*** are cleaned according to user's manual. 3.2 ***Automotive tools and attachments*** are checked for serviceability according to workplace and manufacturers procedures. 3.3 ***Defects and damages*** are reported to immediate superior following industry standards. 3.4 ***Automotive tools and attachments*** are stored according to workplace procedures. 3.5 Safety practices are applied following OSHS. 3.6 ***Wastes*** are disposed following environmental law and regulations.	● Different types of power tools and hand tools ● Techniques in tool Arrangement ● Fundamentals of automotive tools ● Cleaning of automotive tools ● Labeling and arranging of power tools and hand tools ● Safety practices ● Procedures in maintaining of power tools and hand tools ● Tagging of damaged/worn power tools and hand tools ● Reporting damage power tools and hand tools ● Proper disposal of damaged tools ● Proper disposal of chemicals used for cleaning ● OSHS ● Environmental law and regulations ● 5S of good housekeeping ● 3Rs	● Sorting of tools ● Skills in creating reports ● Cleaning of tools ● Checking, cleaning and storing automotive tools and attachments ● Reporting defects and damages ● Disposing wastes ● Practicing safety procedures

RANGE OF VARIABLES

VARIABLE	RANGE
1. Automotive tools	May include: 1.1 Power tools 1.1.1 Electric power tools 1.1.1.1 Electric drill 1.1.2 Pneumatic tools 1.2 Basic tools 1.3 Special service tools (SST)
2. Power sources	May include: 2.1 Electric source 2.2 Pneumatic or air 2.3 Hydraulic
3. Basic tools	May include: 3.1 Wrenches 3.2 Pliers 3.3 Screw drivers 3.4 Power handle 3.5 Ratchet 3.6 Multi-meter tester 3.7 Flash light 3.8 Rubber mallet 3.9 Hammer 3.10 Jack 3.11 Jack stand 3.12 Choke
4. Attachments	May include: 4.1 Bits 4.2 Sockets 4.3 Extension
5. Defects and damages	May include: 5.1 Tools 5.1.1 Cracks 5.1.2 Breakage 5.1.3 Deformity 5.1.4 Looseness 5.1.5 Corrosions 5.1.6 Leaks 5.2 Attachments 5.2.1 Cracks 5.2.2 Breakage 5.2.3 Deformity 5.2.4 Looseness 5.2.5 Corrosions
6. Personal protective equipment (PPEs)	May include: 6.1 Goggles 6.2 Gloves 6.3 Hard hat 6.4 Safety shoes

VARIABLE	RANGE
	6.5 Dust mask
7. Wastes	May include: 7.1 Dead batteries 7.2 Deformed, cracked, broken bits/sockets/extensions 7.3 Used cleaning chemicals 7.4 Used oil 7.5 Contaminated cleaning materials

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment require evidence that the candidate understands the applications and guidelines specified by the manufacturer. 1.1 Prepared automotive tools 1.2 Used Power tools 1.3 Used Hand tools 1.4 Maintained and stored automotive tools 1.5 Disposed wastes 1.6 Applied safety measures
2. Resource implication	The following resource MUST be provided. 2.1 Appropriate power tools and hand tools 2.2 Tools and materials relevant for training 2.3 Proper place for storage and disposal 2.4 Work shop manuals
3. Method of assessment	Competency MUST be assessed through. 3.1 Written examination 3.2 Demonstrations with oral questioning 3.3 Direct observation 3.4 Third party report 3.5 Interview
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : **PERFORM MENSURATION AND CALCULATION**

UNIT CODE : **ALT723214**

UNIT DESCRIPTOR : This unit covers the knowledge and skills on how to use automotive measuring tools.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Select measuring instruments	1.1 Component to be measured is identified based on job requirements. 1.2 ***Automotive measuring instrument*** is identified based on job requirements. 1.3 Correct specifications are obtained from repair manual. 1.4 Measuring tools are calibrated in line with job requirements. 1.5 Measuring instruments are checked for accuracy and adjusted according to manufacturer's manual 1.6 Defective measuring instruments are reported and returned to toolkeeper following industry standards 1.7 Safety practices are applied following OSHS	● Category of measuring instruments ● Types and uses of measuring instruments ● Shapes and Dimensions ● Use of user's manual ● Workshop procedures in reporting defective instruments ● Characteristics of defective measuring instruments ● Procedure in preparing report ● OSHS in calibrating measuring instruments ● Calibration of measuring tools ● Inspection of measuring tools ● Segregation and reporting of defective measuring instruments	● Identifying and selecting measuring instruments ● Visualizing objects and shapes ● Calibration skills ● Identifying defective measuring instruments ● Reporting skills ● Applying safety practices ● Obtaining correct specifications ● Checking measuring instruments for accuracy ● Reporting and segregating defective measuring instruments
2. Carry out measurements and calculation	2.1 ***Automotive measuring instrument*** is selected to achieve required outcome in line with job requirements	● Formulas for volume, areas, perimeters of plane and geometric figures	● Performing calculation ● Applying formulas for volume, areas, perimeters of plane and geometric figures

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.2 Accurate measurements are obtained in line with job requirements 2.3 ***Calculation*** needed to complete work tasks are performed using ***mathematical operations***. 2.4 Numerical computation is self-checked and corrected for accuracy following manufacturer's workshop manual 2.3 Tools' limit of accuracy are read following manufacturer's workshop manual 2.4 Report is submitted to immediate supervisor following industry standard operating procedure 2.5 Safety practices are applied following OSHS	• Different automotive measuring instruments • Calculation & measurement • Four fundamental operation • Linear measurement • Dimensions • Unit conversion • Ratio and proportion • Handling of measuring instruments • Tools' limit of accuracy • OSHS • PPEs	• Handling measuring instruments • Selecting automotive measuring instruments • Obtaining accurate measurements • Performing calculation • Self-checking and correcting numerical computation • Reading tools' limit of accuracy • Applying OSHS • Wearing of PPEs
3. Maintain measuring instruments	3.1 Measuring instruments are handled following manufacturer's manual 3.2 Measuring instruments are cleaned following manufacturer's manual. 3.3 Instruments are stored according to manufacturer's specifications and standard operating procedures. 3.4 Safety practices are applied	• Types of measuring instruments and their uses • Safe handling procedures in using measuring instruments • Four fundamental operation of mathematics • Formula for volume, area, perimeter and other geometric figures • 5S of good housekeeping • Waste management	• Handling and maintaining measuring instruments • Disposing wastes • Practicing good housekeeping • Applying safety practices

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		● Storing of measuring instruments ● OSHS	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Automotive measuring instruments	May include: 1.1 Torque wrench 1.2 Vernier caliper 1.3 Micrometer (inside and outside) 1.4 Dial gauge 1.5 Feeler gauge 1.7 Pitch/thread gauge 1.8 Multi-tester (analog/digital) 1.9 Vacuum Gauge 1.10 Tire depth gauge 1.11 Battery tester 1.12 Steel tape 1.13 Ruler
2. Calculation	May include: 2.1 Volume 2.2 Area 2.3 Displacement 2.4 Inside diameter 2.5 Circumference 2.6 Length 2.7 Thickness 2.8 Outside diameter 2.9 Taper 2.10 Out of roundness 2.11 Voltage 2.12 Resistance 2.13 Current 2.14 Pressure 2.15 Clearance 2.16 Distortion/run-out 2.17 Torque conversion 2.18 Temperature
3. Mathematical operations	Includes: 3.1 Addition 3.2 Subtraction 3.3 Multiplication 3.4 Division 3.5 Fractions 3.6 Percentages 3.7 Mixed numbers

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate perform the following: 1.1 Selected measuring instruments 1.2 Performed measurements and calculation 1.3 Maintained measuring instruments 1.4 Applied safety practices
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Automotive Measuring Tools & equipment 2.3 Materials relevant to the activity 2.4 Training vehicle or simulators 2.5 User's manual 2.6 Repair manual
3. Method of assessment	Competency MUST be assessed through: 3.1 Written exam 3.2 Demonstration with oral questioning 3.3 Third party report 3.4 Interview
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : UTILIZE WORKSHOP FACILITIES AND EQUIPMENT

UNIT CODE : ALT723215

UNIT DESCRIPTOR : This unit deals with inspecting and cleaning of work area including tools, equipment and facilities. Storage of equipment, including operating of basic workshop equipment.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Perform pre-operation activities	1.1 ***Workshop facilities*** are prepared according to work requirements. 1.2 ***Equipment*** are prepared according to work requirements. 1.3 ***Equipment*** are calibrated following users' manual. 1.4 ***Minor repairs*** are carried out based on users' ***manual***. 1.5 Defective equipment are reported to immediate supervisor following company procedures. 1.6 Safety practices are applied following OSHS.	● Different areas of an automotive service facilities. ● Preparation procedures of automotive service facilities ● Different equipment in the automotive service facilities ● Preparation procedures of automotive equipment ● Minor repairs of automotive equipment ● Report of defective equipment ● Reporting procedures for defective equipment ● OSHS practices related to the preparation of facilities and equipment ● Workshop facilities and equipment	● Preparing work area ● Preparing equipment ● Calibrating equipment ● Repairing minor equipment issues ● Reporting defective equipment ● Applying safety practice ● Following manuals

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Use facilities and equipment	2.1 Equipment is operated according to operation ***manual***. 2.2 Facilities are utilized according to workshop procedures. 2.3 Equipment performance is monitored following users' ***manual***. 2.4 Facilities functionalities are monitored following workplace procedures. 2.5 Safety practices are applied following OSHS.	● Operate Equipment ● Identify facilities required for task ● Evaluate equipment operation ● Inspect facility functionalities ● OSHS practices related to operation of facilities and equipment ● Manuals in utilizing facility and equipment ● Monitoring procedure of equipment's performance ● Evaluate equipment operation ● Inspection of facility functionalities	● Operating equipment ● Utilizing facility ● Monitoring equipment performance ● Monitoring functionalities of facility ● Practicing safety ● Following manual
3. Conduct post-operation activities	3.1 Workshop facilities are restored according to 5S of good housekeeping. 3.2 ***Equipment*** are cleaned and stored according to good housekeeping. 3.3 Wastes are disposed following waste management procedure and OSHS. 3.4 ***PPEs*** and Safety practices are applied following OSHS. 3.5 Report is prepared based on workshop procedure.	● 5S of Good housekeeping ● 3Rs/ Waste segregation and disposal ● Restoration of the facilities ● Maintenance and storage of Equipment ● OSHS ● Preparation of report	● Restoring workshop facilities properly ● Cleaning Equipment ● Storing equipment in proper location ● Disposing waste materials ● Reporting facilities and equipment condition ● Practicing safety ● Practicing 5S and 3Rs

RANGE OF VARIABLES

VARIABLE	RANGE
1. Equipment	May include: 1.1 Lifter (Two Post Lifter / Four Post Lifter/ Scissor type) 1.2 Crocodile Jack 1.3 Jack Stand 1.4 Air Compressor 1.5 Oil drain
2. Workshop facilities	May include: 2.1 Service Stall / Working Bay / Workshop areas for servicing/repairing light and/or heavy vehicle and/or plant transmissions and/or outdoor power equipment 2.2 Overhauling Room 2.3 Electrical / Air-con Room 2.4 Inspection Area 2.5 Open workshop/garage and enclosed, ventilated office area 2.6 Car wash area 2.7 Other variables may include workshop with: 2.7.1 Mess hall 2.7.2 Wash room 2.7.3 Comfort room 2.7.4 Storage Room 2.7.5 Training Room
3. Manuals	May include: 3.1 Vehicle/plant manufacturer specifications 3.2 Company operating procedures 3.3 Industry/Workplace Codes of Practice 3.4 Product manufacturer specifications 3.5 Industry Occupational Health &Safety 3.6 Equipment Operation Guidelines 3.7 Service/workshop/repair manual
4. PPEs	May include: 4.1 Gloves 4.2 Apron 4.3 Goggles 4.4 Safety shoes 4.5 Uniforms 4.6 Cap 4.7 Safety helmet
5. Minor repairs	May include: 3.1 Lubrication 3.2 Bolt tightening 3.3 Worn-out parts replacement

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Performed pre-operation activities 1.2 Used facilities and equipment 1.3 Conducted post-operation activities 1.4 Applied safety practices and good housekeeping 1.5 Disposed wastes
2. Resource implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Equipment 2.3 Materials relevant to the activity 2.4 Manuals/references 2.5 PPEs 2.6 Fire Extinguishers
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Written exam 3.2 Demonstration with oral questioning 3.3 Direct observation
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : PREPARE SERVICING PARTS AND CONSUMABLES

UNIT CODE : ALT723216

UNIT DESCRIPTOR : This unit of competency covers the ability to prepare parts and consumables for gasoline and diesel engines in conducting preventive maintenance.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify parts and consumables	1.1 ***Parts and consumables*** are ***determined*** according to job requirements. 1.2 Availability of parts and consumables are confirmed based on stock. 1.3 ***Indirect materials*** are identified according to job requirements. 1.4 ***Hazardous parts and consumables*** are identified according to International standards. 1.5 Safety practices are applied according to OSHS.	● Job requirements ● Safety practices ● Understanding manuals ● Hazardous parts and consumables ● Solid waste management act (RA 6969) ● Wearing of PPE's ● OSHS ● Proper storage of materials ● Chemical contents of consumables ● Composition of consumables ● Quality of parts and consumables ● Computation for quantity of parts and consumables ● Vehicle specifications ● Identifying Part no. ● Awareness in part number ● Updated type of parts and consumables	● Determining parts and consumables ● Reading and interpreting job requirements ● Identifying required parts & consumables ● Understanding safety practices ● Determining quantity and quality of parts and consumables ● Confirming availability of parts and consumables ● Identifying indirect materials ● Identifying hazardous parts and consumables ● Applying safety practices ● Understanding safety practices ● Following manuals
2. Retrieve and withdraw parts and consumables	2.1 Requisition slip is prepared according to identified parts and consumables. 2.2 Withdrawal of parts and materials are recorded. 2.3 Quantity of parts and consumables	● Job requirements ● Safety practices ● Understanding manuals ● Hazardous parts and consumables ● Solid waste management act (RA 6969)	● Reading and interpreting requisition slip ● Validating quantity of parts and materials ● Handling parts and consumables

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	are validated according to job requirements 2.4 Parts and materials are handled following safety procedures.	● Wearing of PPE's ● Updated types of parts & consumables for proper usage	
3. Complete work process	3.1 Used parts and consumables are labeled and segregated 3.2 Used parts are packed and returned to customers 3.3 Consumables are collected for recycling 3.4 PPEs are worn following OSHS. 3.5 ***Wastes*** are disposed according to RA 6969.	● Labeling and segregation of used parts and consumables ● Job requirements ● Safety practices ● 3Rs ● Solid waste management act (RA 6969) ● Wearing of PPE's	● Waste segregation and disposal of parts & consumables according to RA 6969

RANGE OF VARIABLES

VARIABLE	RANGE
1. Parts and consumables	May include: 1.1 Engine oil 1.2 Clutch fluid 1.3 Transmission oil 1.4 Differential oil 1.5 Power steering fluid 1.6 Brake fluid 1.7 Engine coolant 1.8 Engine oil filter 1.9 Fuel filter 1.10 Air cleaner element 1.11 Feed pump strainer 1.12 Sparkplugs (Gasoline engine) 1.13 Battery 1.14 Air cleaner 1.15 Tire 1.16 Wiper blade 1.17 A/C pollen filter 1.18 Bulb 1.19 Brake pad/brake shoe 1.20 Clutch lining
2. Determining parts and consumables	May include: 2.1 Quantity 2.2 Quality
3. Indirect materials	May include: 3.1 Rags 3.2 Saw dust 3.3 Cleaning fluids 3.4 Sand paper
4. Hazardous parts consumables	May include: 4.1 Batteries 4.2 Used oil 4.3 Used fluids 4.4 Used coolant 4.5 Used parts 4.6 Used oil filter
5. Wastes	May include: 5.1 Contaminated consumables 5.2 Contaminated parts

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Identified parts and consumables 1.2 Retrieved and withdrawn parts and consumables 1.3 Completed work process 1.4 Applied safety practices
2. Resource implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Materials relevant to the activity 2.3 Repair manuals and related reference materials
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Direct observation 3.2 Interview 3.3 Written examination 3.4 Demonstration with oral questioning 3.5 Third party report
4. Context of Assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : PREPARE VEHICLE FOR SERVICING AND RELEASING

UNIT CODE : ALT723217

UNIT DESCRIPTOR : This unit covers the knowledge, skills, and attitudes needed in identifying and preparing the vehicle for servicing and releasing.

ELEMENT	**PERFORMANCE CRITERIA** ***Italicized terms*** are elaborated in the Range of Variables	**REQUIRED KNOWLEDGE**	**REQUIRED SKILLS**
1. Receive vehicle	1.1 Vehicle is located following company standard. 1.2 Checklist is validated for exterior and interior items in accordance with ***vehicle checklist***. 1.3 Job Order is checked for proper assignment according to ***work classification***. 1.4 ***Work bay*** for vehicle is designated based from Job Order. 1.5 Vehicle is moved on the designated ***work bay***.	● Identification of basic vehicle components ● Types of defects ● Read & understand Job Order ● Flat rate time ● Use of PPEs ● Adherence to safety procedures ● Vehicle checklist ● Work classification ● Work bay **Attitudes:** ● Patient ● Attention to details ● Honest ● Time Conscious	● Completing vehicle checklist ● Classifying work to be performed ● Assigning work bay ● Validating checklist for exterior and interior items ● Checking job order for proper assignment ● Identifying vehicle ● Moving vehicle to designated work bay
2. Prepare vehicle for servicing	2.1 ***Protective covers*** are installed prior to servicing based on workshop operating standards 2.2 Vehicle is positioned and set-up for lifting according to repair order. 2.3 Vehicle is lifted for servicing following manufacturer's manual. 2.4 Safety practices are applied following safety procedures.	● Familiarization on equipment & facilities ● Time estimation of completion ● Vehicle tagging ● Types of protective covers ● Setting-up of vehicle for lifting ● Read & understand repair order ● Use of PPEs ● Use of safety gears ● OSHS ● Adherence to safety procedures	● Understanding of vehicle status ● Installation of protective covers ● Positioning vehicle ● Operating lifter ● Moving vehicle ● Setting-up vehicle for lifting ● Practicing safety

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		Attitudes: ● Patient ● Attention to details ● Honest ● Time Conscious	
3. Prepare vehicle for releasing	3.1 Job done is confirmed according to repair order. 3.2 Quality check is done based from repair order. 3.3 Transfer of vehicle to wash bay is coordinated according to SOP. 3.3 Vehicle is endorsed to quality control person following workplace procedure.	● Familiarization of equipment & facilities ● Read & understand repair order ● Confirmation of job done ● Quality standards checking ● Coordination of transferring vehicle ● Endorsement procedures for vehicle **Attitudes:** ● Patient ● Attention to details ● Honest ● Time Conscious	● Confirming job done ● Performing quality checking ● Coordinating transfer of vehicle to wash bay ● Endorsing and turning-over vehicle

RANGE OF VARIABLES

VARIABLE	RANGE
1. Vehicle checklist	May include: 1.1 External scratches, accessories, items, dents, damages and cracks 1.2 Internal items, scratches, noticeable damages, including spare tire, tools, and loose items 1.3 Standard items that are not present during inspection 1.4 Valuable/personal belongings
2. Work classification	May include: 2.1 Body and Paint repair 2.2 General Job repair 2.3 Periodic maintenance service (PMS)
3. Work bay	May include: 3.1 Service Stall / Working Bay / Workshop areas for servicing/repairing light and/or heavy vehicle and/or plant transmissions and/or outdoor power equipment 3.2 Overhauling Room 3.3 Electrical / Air-con Room 3.4 Inspection Area 3.5 Open workshop/garage and enclosed, ventilated office area
5. Protective covers	May include but not limited to: 5.1 Seat Cover 5.2 Steering Wheel Cover 5.3 Handbrake Cover 5.4 Shift Knob Cover 5.5 Fender Cover 5.6 Paper mat

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Received vehicle 1.2 Prepared vehicle for servicing 1.3 Prepared vehicle for releasing 1.4 Applied safety practices
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & Equipment 2.3 Materials relevant to the activity 2.4 Manuals and references
3. Method of assessment	Competency may be assessed through: 3.1 Direct observation 3.2 Demonstration with Oral questioning 3.3 Interview 3.4 Written Evaluation 3.5 Third Party Report
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

CORE COMPETENCIES

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR PETROL ELECTRONIC FUEL INJECTION SYSTEM

UNIT CODE : ALT723381

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair petrol electronic fuel injection system.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose electronic fuel injection system	1.1 ***Job requirements*** are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed following ***industry criteria***. 1.4 ***Tools and equipment*** are selected and checked for serviceability according to ***industry criteria***. 1.5 Defective ***tools and equipment*** are reported following workplace procedures. 1.6 ***Parts and supplies*** are sourced following workplace procedures.	1.1 OSHS 1.2 Trouble shooting guide. 1.3 Waste management 1.4 Sourcing out and interpretation of repair information 1.5 Service/Repair manual 1.6 Tools, equipment, parts and supplies in inspecting and repairing electronic fuel injection system. 1.7 Different job requirements 1.8 Serviceability of tools and equipment 1.9 Work hazards 1.10 Repair Order checklist 1.11 Identification and function of electronic fuel injection system.	1.1 Determining job requirements from workplace instructions 1.2 Sourcing and interpreting diagnostic information 1.3 Identifying hazards 1.4 Managing hazard risks 1.5 Checking software updates 1.6 Clarifying instructions 1.7 Locating appropriate sources of information 1.8 Selecting and checking tools and equipment 1.9 Reporting defective tools and equipment 1.10 Sourcing of parts and supplies 1.11 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	1.7 Safety practices are applied following OSHS.		
2. Perform diagnosis	2.1 ***Diagnostic procedures*** are performed following ***industry criteria***. 2.2 Diagnostic symptoms are analyzed using troubleshooting guide and most appropriate to the circumstances. 2.3 ***Diagnostic tools*** are operated following manufacturer's specification. 2.4 Inspection findings and recommendations are reported according to manufacturer specification workshop procedure and ***industry criteria***. 2.5 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Interpretation of job requirements 2.2 Identification and function of electronic fuel injection system. 2.3 Trouble shooting guide 2.4 Related system analysis 2.5 Mensuration 2.6 Arithmetic operations 2.7 Use of measuring devices 2.8 Operate diagnostic tools 2.9 Preparation of report on inspection findings, recommendation and repair instructions 2.10 Diagnostic procedures 2.11 OSHS 2.12 Wearing of PPEs 2.13 Industry criteria	2.1 Performing diagnostic procedures 2.2 Interpreting and comparing diagnostic data using trouble shooting guide and manufacturer's manual 2.3 Inspecting electronic fuel injection system. 2.4 Identifying faults and its cause 2.5 Reporting inspection findings, recommendation and repair instructions 2.6 Applying safety practices 2.7 Mensuration skills 2.8 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair electronic fuel injection system	3.1 ***Repairs*** are carried out according to manufacturer specification workshop procedure and ***industry criteria***. 3.2 ***Functionality test*** is carried out according to workplace procedures. 3.3 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.4 Documentation report is prepared based on analysis result. 3.5 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	3.1 Different repairs for electronic fuel injection system 3.2 Identification and function of electronic fuel injection system. 3.3 Arithmetic operations 3.4 Mensuration 3.5 Use of measuring devices 3.6 Service/Repair Manual 3.7 Functionality test for electronic fuel injection system. 3.8 Isolation and elimination approach 3.9 Waste management 3.10 OSHS 3.11 Wearing of PPEs 3.12 Accomplishment of Repair Order	3.1 Sourcing of information 3.2 Interpreting information from manufacturer and workshop literature 3.3 Applying safety practices. 3.4 Mensuration skills 3.5 Applying arithmetic operations 3.6 Repairing electronic fuel injection system. 3.7 Performing functionality test 3.8 Accomplishing Repair Order 3.9 Analyzing and selecting repair options 3.10 Preparing documentation report
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is turned-over to immediate superior for quality control following workplace procedure. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations.	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 Waste Management 4.6.1 3Rs 4.7 5S	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 ***Workplace documents*** are accomplished according to workplace procedures.	4.8 Checking and storage of tools and equipment 4.9 Workplace documents	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Inspect and replace fuel injector 1.2 Inspect and replace fuel rail 1.3 Inspect and replace fuel pressure regulator 1.4 Inspect and replace fuel pulsation damper 1.5 Inspect fuel filter 1.6 Inspect and replace electric fuel pump 1.7 Inspection of fuel lines /piping 1.8 Inspect and replace engine control system components
2. Industry criteria	Industry criteria may include: 2.1 Repair manual 2.2 Workplace procedures 2.3 Safety and environmental requirements 2.4 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Fuel pressure gauge 3.1.4 Multi tester (analog) 3.1.5 Multi tester (digital) 3.1.6 Trouble light 3.1.7 Soldering kit 3.1.8 Graduated cylinder 3.2 Equipment: 3.2.1 Lifter 3.2.2 Diagnostic Scan tools with oscilloscope
4. Parts and supplies	Parts and supplies may include: 4.1 Parts (included in the simulator component or training car) 4.1.1 Electric fuel pump 4.1.2 Fuel pressure regulator 4.1.3 Fuel Filter 4.1.4 Fuel pulsation damper 4.1.5 Fuel rail 4.1.6 Fuel injector 4.1.7 ECU – electronic control unit 4.1.8 Shrinkable tube 4.1.9 Cable tie 4.2 Supplies: 4.2.1 Electrical tape 4.2.2 Rubber O-ring 4.2.3 Contact cleaners 4.2.4 Back probe

VARIABLE	RANGE
	4.2.5 Soldering kit 4.2.6 Shrinkable tube 4.2.7 Cable tie 4.2.8 Rags 4.2.9 Vehicle protective kit 4.2.9.1 Seat cover 4.2.9.2 Steering wheel cover 4.2.9.3 Fender cover 4.2.9.4 Shifting knob / lever cover 4.2.9.5 Paper matting 4.2.10 PPEs
5. Diagnostic procedures	Diagnostic procedures may include: 5.1 Manual diagnosis (symptoms and problems) 5.2 Diagnostic scan tool (diagnostic trouble code-DTC)
6. Diagnostic tools	Diagnostic tools may include: 6.1 Scan tool 6.2 Multi-tester
7. Repair of electronic fuel injection system	Repair of electronic fuel injection system may include: 7.1 Removal and installation of fuel rail assembly 7.2 Replacement of defective parts of fuel pressure regulator 7.3 Checking and cleaning of fuel line 7.4 Replacement of fuel filter 7.5 Removal and replacement of fuel pump assembly 7.6 Soldering and reconnecting wiring harness 7.7 Replacement of connectors and terminals
8. Functionality test	Functionality test may include: 8.1 Inspection for fuel lines for leaks 8.2 Inspection of fuel pump 8.3 Check for vibration and noise 8.4 Checking and analyzing data list
9. Workplace documents	Workplace documents may include: 9.1 Repair order 9.2 Inspection form

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose electronic fuel injection system. 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards associated with work and managed risks. 1.1.4 Selected and checked serviceability of tools and equipment. 1.1.5 Reported defective tools and equipment. 1.1.6 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Performed diagnostic procedures. 1.2.2 Analyzed diagnostic symptoms. 1.2.3 Operated diagnostic tools. 1.2.4 Reported inspection findings and recommendations. 1.2.5 Applied safety practices. 1.3 Repaired electronic fuel injection system. 1.3.1 Carried out repairs. 1.3.2 Carried out functionality test. 1.3.3 Analyzed and selected repair options and solutions. 1.3.4 Prepared documentation report. 1.3.5 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Turned-over vehicle. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Accomplished workplace documents.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Tools, materials & equipment relevant to perform required tasks 2.3 Manufacturer's repair manual 2.4 PPEs 2.5 Training vehicle 2.6 First-aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration with Oral questioning 3.2 Written exam 3.3 Direct Observation
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR EMISSION CONTROL SYSTEM

UNIT CODE : ALT723399

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair emission control system of petrol and diesel engine including evaporative and exhaust gas recirculation.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair vehicle emission control system	1.1 ***Job requirements*** are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed following ***industry criteria***. 1.4 Tools and equipment are selected and checked for serviceability according to ***industry criteria***. 1.5 Defective ***tools and equipment*** are reported following workplace procedures. 1.6 Availability of ***materials, parts and supplies*** are checked and reported following workplace procedures.	1.1 OSHS 1.2 Trouble shooting guide. 1.3 Waste management 1.4 Sourcing out and interpretation of repair information 1.5 Service/Repair manual 1.6 Tools, equipment materials, parts and supplies in inspecting and repairing faults on emission control system 1.7 Different job requirements 1.8 Serviceability of tools and equipment 1.9 Work hazards 1.10 Repair Order checklist 1.11 Identification and function of emission control system.	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Identifying hazards associated with work 1.5 Selecting and checking tools and equipment 1.6 Reporting defective tools and equipment 1.7 Checking and reporting the availability of materials, parts and supplies 1.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	1.7 Safety practices are applied following OSHS.		
2. Perform diagnosis	2.1 ***Diagnostic procedures*** are performed following ***industry criteria***. 2.2 Diagnostic symptoms are analyzed using troubleshooting guide and most appropriate to the circumstances. 2.3 Inspection findings and recommendations are reported according to workplace procedure and ***industry criteria***. 2.4 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Interpretation of job requirements 2.2 Identification and function of emission control system. 2.3 Trouble shooting guide 2.4 Related system analysis 2.5 Mensuration 2.6 Arithmetic operations 2.7 Diagnostic procedures 2.8 Use of measuring devices 2.9 Operate diagnostic tools 2.10 Reporting procedures 2.11 Preparation of report on inspection findings, recommendation and repair instructions 2.12 Awareness on the use of opacity meter 2.13 OSHS 2.14 Wearing of PPEs 2.15 Industry criteria	2.1 Performing diagnostic procedures 2.2 Interpreting and comparing diagnostic data using trouble shooting guide and manufacturer's manual 2.3 Inspecting emission control system 2.4 Reporting inspection findings, recommendation and repair instructions 2.5 Analyzing inspection results 2.6 Reporting and recommendation s inspection findings 2.7 Applying safety practices 2.8 Mensuration skills 2.9 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair emission control system	3.1 Exhaust gas recirculation is removed following ***industry criteria***. 3.2 ***Repairs*** are carried out according to ***industry criteria***. 3.3 ***Functionality test*** is carried out according to workplace procedures. 3.4 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.5 Documentation report is prepared based on analysis result. 3.6 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	3.1 Different repairs for emission control system 3.2 Identification and function of emission control system 3.3 Arithmetic operations 3.4 Mensuration 3.5 Use of measuring devices 3.6 Service/Repair Manual 3.7 Functionality test for electronic fuel injection system. 3.8 OSHS 3.9 Wearing of PPEs 3.10 Accomplishment of Repair Order 3.11 Isolation and elimination approach 3.12 Waste management	3.1 Sourcing of information 3.2 Interpreting information from manufacturer and workshop literature 3.3 Applying safety practices 3.4 Applying arithmetic operations 3.5 Repairing emission control system 3.6 Performing functionality test 3.7 Accomplishing Repair Order 3.8 Analyzing and selecting repair options 3.9 Preparing documentation report
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is turned-over to immediate superior for quality control following workplace procedure. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations.	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 5S 4.7 Waste management 4.7.1 3Rs	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Performing and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 ***Workplace documents*** are accomplished according to workplace procedures.	4.8 Checking and storage of tools and equipment 4.9 Workplace documents	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Check and replace EGR valve 1.2 Check and replace positive crank case ventilation (PCV) valve 1.3 Replacement of fuel filter 1.4 Replacement of oxygen sensor 1 1.5 Replacement of oxygen sensor 2 1.6 Replacement of Catalytic converter 1.7 Check and replace solenoid valve 1.8 Inspect exhaust gas recirculation 1.9 Inspect and replace exhaust gas recirculation pipe 1.10 Inspect and replace exhaust gas recirculation motor 1.11 Inspect EGR pipe for leaks 1.12 Inspect exhaust manifold for leaks 1.13 Inspect EGR gasket 1.14 Inspect and replace catalytic converter 1.15 Positive crankcase ventilation valve 1.16 Inspect and replace engine control module 1.17 Inspect and replace charcoal canister 1.18 Check and analyze exhaust gas
2. Industry criteria	Industry criteria may include: 2.1 Repair manual 2.2 Workplace procedures 2.3 Safety and environmental requirements 2.4 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Trouble light 3.1.4 Vacuum tester 3.1.5 Multi tester 3.1.6 Test harness 3.2 Equipment: 3.2.1 Lifter 3.2.2 Diagnostic scan tool with oscilloscope 3.2.3 Emission tester
4. Materials, parts and supplies	Materials, parts and supplies may include: 4.1 Materials: 4.1.1 Rags 4.1.2 Grease 4.1.3 Penetrating oil 4.1.4 PPEs 4.2 Supplies: 4.2.1 Thinner/Solvent/ kerosene

VARIABLE	RANGE
	4.2.2 Sand paper 4.2.3 Air compressor 4.3 Parts: 4.3.1 EGR valve 4.3.2 PCV valve 4.3.3 Fuel filter 4.3.4 Oxygen sensor 1 4.3.5 Oxygen sensor 2 4.3.6 Catalytic converter 4.3.7 Gasket 4.3.8 Solenoid valve
5. Diagnostic procedures	Diagnostic procedures may include: 5.1 Manual diagnosis (symptoms and problems) 5.2 Diagnostic scan tool (diagnostic trouble code-DTC)
6. Repair of emission control system	Repair of emission control system may include: 6.1 Removal and installation of exhaust gas recirculation 6.2 Replacement of catalytic converter 6.3 Removal and installation of oxygen sensor 6.4 Cleaning of exhaust pipe 6.5 Removal and replacement of EGR valve 6.6 Checked for leaks fuel lines
7. Functionality test	Functionality test may include: 7.1 Inspection for exhaust gas recirculation for leaks and clogged 7.2 Inspection of crankcase ventilation for leaks and clogged 7.3 Check for vibration and noise 7.4 check emission reading within standard 7.5 Checking and analyzing data list
8. Workplace documents	Workplace documents may include: 8.1 Repair order 8.2 Inspection form

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair vehicle emission control system. 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards associated with work and managed risks. 1.1.4 Selected and checked serviceability of tools and equipment. 1.1.5 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Performed diagnostic procedures. 1.2.2 Analyzed diagnostic symptoms. 1.2.3 Reported inspection findings and recommendations. 1.2.4 Applied safety practices. 1.3 Repaired emission control system. 1.3.1 Removed exhaust gas recirculation. 1.3.2 Carried out repairs. 1.3.3 Carried out functionality test. 1.3.4 Analyzed and selected repair options and solutions. 1.3.5 Prepared documentation report. 1.3.6 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Turned-over vehicle. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Accomplished workplace documents.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & Equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration with Oral questioning 3.2 Written exam 3.3 Direct Observation
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR DIESEL ELECTRONIC FUEL INJECTION SYSTEM

UNIT CODE : ALT7233100

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair diesel electronic fuel injection (EFI) and common- rail direct injection (CRDI) except petrol electronic fuel injection (EFI).

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair diesel fuel injection system and common-rail EFI fuel injection system	1.1 ***Job requirements*** are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed following ***industry criteria***. 1.4 ***Tools and equipment*** are selected and checked for serviceability according to ***industry criteria***. 1.5 Defective ***tools and equipment*** are reported following workplace procedures. 1.6 ***Materials, parts and supplies*** are source following workplace procedures. 1.7 Safety practices are applied following OSHS.	1.1 OSHS 1.2 Trouble shooting guide. 1.3 Waste management 1.4 Sourcing out and interpretation of repair information 1.5 Service/Repair manual 1.6 Tools, equipment materials, parts and supplies in inspecting and repairing faults on fuel injection system and common-rail EFI fuel injection system. 1.7 Different job requirements 1.8 Serviceability of tools and equipment 1.9 Work hazards 1.10 Repair Order checklist 1.11 Identification and function of diesel fuel injection system and common-rail EFI fuel injection system	1.1 Determining job requirements from workplace instructions 1.2 Sourcing and interpreting diagnostic information 1.3 Identifying hazards 1.4 Managing hazard risks 1.5 Checking software updates 1.6 Clarifying instructions 1.7 Locating appropriate sources of information 1.8 Selecting and checking tools and equipment 1.9 Reporting defective tools and equipment 1.10 Sourcing of materials, parts and supplies 1.11 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Perform diagnosis	2.1 ***Diagnostic procedures*** are performed following ***industry criteria***. 2.2 ***Diagnostic tools*** are operated following manufacturer's specification 2.3 Diagnostic symptoms for ***diesel electronic fuel injection system*** are analyzed using troubleshooting guide and most appropriate to the circumstances. 2.4 Inspection findings and recommendations are reported according to ***industry criteria***. 2.5 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Interpretation of job requirements 2.2 Diagnostic procedures 2.3 Diagnose and repair common-rail EFI fuel injection system. 2.4 Diagnose and repair diesel fuel injection system 2.5 Identification and function of fuel injection system and common-rail EFI fuel injection system. 2.6 Trouble shooting guide 2.7 Related system analysis 2.8 Mensuration 2.9 Arithmetic operations 2.10 Use of measuring devices 2.11 Operate diagnostic tools 2.12 Preparation of report on inspection findings, recommendation and repair instructions 2.13 OSHS 2.14 Wearing of PPEs 2.15 Industry criteria	2.1 Interpreting and comparing diagnostic data using trouble shooting guide and manufacturer's manual 2.2 Performing diagnostic procedures 2.3 Inspecting fuel injection system and common-rail EFI fuel injection system 2.4 Reporting inspection findings, recommendation and repair instructions 2.5 Reporting and recommendation s inspection findings 2.6 Applying safety practices 2.7 Mensuration skills 2.8 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair diesel fuel injection system and common-rail EFI fuel injection system	3.1 ***Repairs*** are carried out according to ***industry criteria***. 3.2 ***Functionality test*** is carried out according to workplace procedures. 3.3 ***Documentation report*** is prepared based on analysis result. 3.4 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	3.1 Different repairs for fuel injection system and common-rail EFI fuel injection system. 3.2 Identification and function of fuel injection system and common-rail EFI fuel injection system 3.3 Arithmetic operations 3.4 Mensuration 3.5 Use of measuring devices 3.6 Service/Repair Manual 3.7 Functionality test for fuel injection system and EFI fuel injection system. 3.8 Isolation and elimination approach 3.9 Waste management 3.10 OSHS 3.11 Wearing of PPEs 3.12 Accomplishment of Repair Order	3.1 Sourcing of information 3.2 Interpreting information from manufacturer and workshop literature 3.3 Identifying faults and its cause 3.4 Applying safety practices 3.5 Applying arithmetic operations 3.6 Repairing fuel injection system. 3.7 Performing functionality test 3.8 Accomplishing Repair Order 3.9 Analyzing and selecting repair options 3.10 Preparing documentation report
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is turned-over to immediate superior for quality control following workplace procedure. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 5S	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 ***Workplace documents*** are accomplished according to workplace procedures.	4.7 Waste management 4.7.1 3Rs 4.8 Checking and storage of tools and equipment 4.9 Workplace documents	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Inspect and replace fuel injector 1.2 Inspect and replace fuel injection and supply pump 1.3 Inspect and replace fuel filter secondary 1.4 Inspect and replace fuel filter primary 1.5 Bleeding of fuel systems as required 1.6 Inspect fuel lines for leaks 1.7 Servicing fuel filters 1.8 Inspect and replace common-rail 1.9 Inspect common-rail system using scanner
2. Industry criteria	Industry criteria may include: 2.1 Repair manual 2.2 Workplace procedures 2.3 Safety and environmental requirements 2.4 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Fuel injector tester 3.1.4 Multi tester (analog) 3.1.5 Multi tester (digital) 3.1.6 Trouble light 3.1.7 Soldering kit 3.2 Equipment: 3.2.1 Lifter 3.2.2 Diagnostic scan tool with oscilloscope
4. Materials, parts and supplies	Materials, parts and supplies may include: 4.1 Materials: 4.1.1 Rags 4.1.2 PPEs 4.2 Supplies: 4.2.1 Soldering lead 4.2.2 Soldering paste 4.3 Parts: 4.3.1 Rubber O –ring 4.3.2 Gasket 4.3.3 Fuel injector 4.3.4 Fuel injection and supply pump 4.3.5 Fuel filter secondary 4.3.6 Fuel filter primary 4.3.7 Fuel filters 4.3.8 Common-rail
5. Diagnostic procedures	Diagnostic procedures may include:

VARIABLE	RANGE
	5.1 Manual diagnosis (symptoms and problems) 5.2 Diagnostic scan tool (diagnostic trouble code-DTC)
6. Diagnostic tools	Diagnostic tools may include: 6.1 Scanner 6.2 Computer with OBD software(diesel) 6.3 Multi tester
7. Diesel electronic fuel injection system	Diesel electronic fuel injection system may include: 7.1 Conventional diesel electronic fuel injection system 7.2 Common-rail direct injection system (CRDI)
8. Repairs of fuel injection system	Repairs of fuel injection system may include: 8.1 Removal and installation of injection and supply pump/ injection pump 8.2 Removal and installation of fuel injector 8.3 Checked for leaks fuel lines 8.4 Removal and installation of fuel injection sensors
9. Functionality test	Functionality test may include: 9.1 Inspection of fuel lines for leaks 9.2 Inspection of tightening injection bracket 9.3 Check for vibration and noise 9.4 Checking and analyzing data list
10. Documentation report	Documentation report includes: 10.1 Diagnostic report 10.2 Recommendation 10.3 Accomplishments
11. Workplace documents	Workplace documents may include: 11.1 Repair order 11.2 Inspection form 11.3 Diagnostic form 11.4 Waste disposal form

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair diesel fuel injection system and common-rail EFI fuel injection system. 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards associated with the work and managed risks. 1.1.4 Selected and checked serviceability of tools and equipment. 1.1.5 Sourced materials, parts and supplies. 1.1.6 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Performed diagnostic procedures. 1.2.2 Operated diagnostic tools. 1.2.3 Analyzed diagnostic symptoms for diesel electronic fuel injection system. 1.2.4 Reported inspection findings and recommendations. 1.2.5 Applied safety practices. 1.3 Repaired diesel fuel injection system and common-rail EFI fuel injection system 1.3.1 Carried out repairs. 1.3.2 Carried out functionality test. 1.3.3 Prepared documentation report. 1.3.4 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Turned-over vehicle. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Accomplished workplace documents.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration with Oral questioning 3.2 Written exam 3.3 Direct Observation

4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR FORCED-INDUCTION SYSTEM

UNIT CODE : ALT7233101

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair of forced induction system specifically turbocharger system.

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair a forced-induction system	1.1 ***Job requirements*** are determined from workplace instructions 1.2 Diagnostic information is sourced and interpreted according to workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed following ***industry criteria***. 1.4 ***Tools and equipment*** are selected and checked for serviceability according to ***industry criteria***. 1.5 Defective tools and equipment are reported following workplace procedures. 1.6 ***Parts and materials*** are sourced following workplace procedures. 1.7 Safety practices are applied following OSHS.	1.1 OSHS 1.2 Trouble shooting guide. 1.3 Diagnose and repair forced induction system. 1.4 Waste management 1.5 Sourcing out and interpretation of repair information 1.6 Service/Repair manual 1.7 Tools, equipment and materials in inspecting and repairing faults on forced induction system. 1.8 Interpretation of job requirements 1.9 Different job requirements 1.10 Serviceability of tools and equipment 1.11 Work hazards 1.12 Accomplishment of Repair Order checklist 1.13 Identification and function forced-induction system	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Selecting and checking tools and equipment 1.5 Reporting defective tools and equipment 1.6 Checking and reporting the availability of materials 1.7 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Diagnose a forced-induction system	2.1 ***Diagnostic procedures*** are performed following ***industry criteria***. 2.2 Diagnostic symptoms for ***forced induction system*** are analyzed using troubleshooting guide and most appropriate to the circumstances. 2.3 Inspection findings and recommendations are reported according to ***industry criteria***. 2.4 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Identification and function forced induction system. 2.2 Diagnostic procedures 2.3 Trouble shooting guide 2.4 Mensuration 2.5 Arithmetic operations 2.6 Use of measuring devices 2.7 Reporting procedures 2.8 OSHS 2.9 Wearing of PPEs 2.10 Industry criteria	2.1 Interpreting information from manufacturer and workplace procedure 2.2 Performing diagnostic procedures 2.3 Inspecting forced induction system 2.4 Comparing inspection results 2.5 Reporting and recommendation s inspection findings 2.6 Applying safety practices 2.7 Mensuration skills 2.8 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair forced-induction system	3.1 Turbocharger assembly is removed following ***industry criteria***. 3.2 Turbocharger is repaired according to ***industry criteria***. 3.3 ***Functionality test*** is carried out according to workplace procedures. 3.4 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.5 Documentation report is prepared based on analysis result. 3.6 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	3.1 Different repairs for forced induction system. 3.2 Identification and forced induction system. 3.3 Arithmetic operations 3.4 Mensuration 3.5 Use of measuring devices 3.6 Service/Repair Manual 3.7 Post repair testing for fuel injection system. 3.8 OSHS 3.9 Wearing of PPEs 3.10 Accomplishment of Repair Order	3.1 Sourcing of information 3.2 Interpreting information from manufacturer and workshop literature. 3.3 Identifying faults and its cause 3.4 Using precision measuring equipment including dial indicator, vacuum, pressure gauges, temperature gauge 3.5 Applying safety practices 3.6 Applying arithmetic operations 3.7 Repairing exhaust gas recirculation system. 3.8 Performing post-repair testing 3.9 Accomplishing Repair Order
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is turned-over to immediate superior for quality control following workplace procedure. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA *Italicized terms* are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 ***Workplace documents*** are accomplished according to workplace procedures.	4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Diagnose and Inspect turbo charger assembly 1.2 Replace intercooler 1.3 Diagnose and Inspect oil supply and return system 1.4 Diagnose and Inspect turbocharger boost mechanism (variable nozzle) 1.5 Diagnose and Inspect turbo hoses 1.6 Diagnose and Inspect exhaust manifold for leaks 1.7 Diagnose and Inspect intake and exhaust blockage 1.8 Diagnose and Inspect induction control system
2. Industry criteria	Industry criteria may include: 2.1 Repair manual 2.2 Workplace procedures 2.3 Safety and environmental requirements 2.4 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Vacuum tester 3.1.4 Trouble light 3.1.5 Dial gauge with magnetic stand 3.1.6 Multi-tester (digital) 3.2 Equipment: 3.2.1 Lifter 3.2.2 Diagnostic scan tool with oscilloscope
4. Parts	Parts may include: 4.1 Turbocharger 4.2 Induction control system 4.3 Intercooler 4.4 Turbo hose
5. Materials	Materials may include: 5.1 Rags 5.2 Grease 5.3 Penetrating oil 5.4 PPEs 5.5 Vehicle protective kit 5.5.1 Seat cover 5.5.2 Steering wheel cover 5.5.3 Fender cover 5.5.4 Shifting knob/ lever cover 5.5.5 Paper matting
6. Diagnostic procedures	Diagnostic procedures may include: 6.1 Manual diagnosis (symptoms and problems) 6.2 Diagnostic scan tool (diagnostic trouble code-DTC)
7. Forced induction system	Forced induction system includes:

VARIABLE	RANGE
	7.1 Conventional turbocharger 7.2 Variable geometric turbo (VGT) charger
8. Functionality test	Functionality test may include: 8.1 Checking and analyzing data list 8.2 Check active/actuator test using diagnostic scan tool 8.3 Test drive
9. Workplace documents	Workplace documents may include: 9.1 Repair order 9.2 Inspection form 9.3 Diagnostic form

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair a forced-induction system 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards associated with the work and managed risks. 1.1.4 Selected and checked tools and equipment. 1.1.5 Reported defective tools and equipment. 1.1.6 Sourced parts and materials. 1.1.7 Applied safety practices. 1.2 Diagnosed a forced-induction system 1.2.1 Performed diagnostic procedures. 1.2.2 Analyzed diagnostic symptoms for forced induction system. 1.2.3 Reported inspection findings and recommendations. 1.2.4 Applied safety practices. 1.3 Repaired forced-induction system 1.3.1 Removed turbocharger assembly. 1.3.2 Repaired turbocharger. 1.3.3 Carried out functionality test. 1.3.4 Analyzed and selected repair options and solutions. 1.3.5 Prepared documentation report. 1.3.6 Applied safety practices. 1.4 Completed work processes 1.4.1 Conducted final inspection. 1.4.2 Turned-over vehicle. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Accomplished workplace documents.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer’s repair manual 2.5 PPEs 2.6 Training vehicle
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration with Oral questioning 3.2 Written exam 3.3 Direct Observation
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR ELECTRONIC SPARK ADVANCE SYSTEM

UNIT CODE : ALT7233102

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair electronic spark advance system.

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair a spark advance system	1.1 ***Job requirements*** are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed following ***industry criteria***. 1.4 ***Tools and equipment*** are selected and checked for serviceability according to ***industry criteria***. 1.5 Defective ***tools and equipment*** are reported following workplace procedures. 1.6 Availability of ***materials,*** parts and supplies are checked and reported following workplace procedures. 1.7 Safety practices are applied following OSHS.	1.1 OSHS 1.2 Trouble shooting guide. 1.3 Diagnose and repair spark advance system. 1.4 Waste management 1.5 Sourcing out and interpretation of repair information 1.6 Service/Repair manual 1.7 Tools, equipment and materials in inspecting and repairing faults on spark advance system. 1.8 Interpretation of job requirements 1.9 Different job requirements 1.10 Serviceability of tools and equipment 1.11 Work hazards 1.12 Accomplishment of Repair Order checklist 1.13 Identification sparks advance system	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Selecting and checking tools and equipment 1.5 Reporting defective tools and equipment 1.6 Checking and reporting the availability of materials 1.7 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Diagnose a spark advance system	2.1 ***Diagnostic procedures*** are performed following ***industry criteria***. 2.2 Diagnostic symptoms are analyzed using troubleshooting guide and most appropriate to the circumstances. 2.3 Inspection findings and recommendations are reported according to ***industry criteria***. 2.4 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Identification and function spark advance system. 2.2 Diagnostic procedures 2.3 Trouble shooting guide 2.4 Mensuration 2.5 Arithmetic operations 2.6 Use of measuring devices 2.7 Use of scan tools 2.8 Reporting procedures 2.9 OSHS 2.10 Wearing of PPEs 2.11 Industry criteria	2.1 Interpreting information from manufacturer and workplace procedure 2.2 Performing diagnostic procedures 2.3 Inspecting forced induction system 2.4 Comparing inspection results 2.5 Reporting and recommendation s inspection findings 2.6 Applying safety practices 2.7 Mensuration skills 2.8 Applying arithmetic operations
3. Repair a spark advance system	3.1 Spark advance system component is removed following ***industry criteria***. 3.2 ***Repair for electronic spark advance system components*** are carried out according to industry criteria. 3.3 ***Functionality test*** is carried out according to workplace procedures. 3.4 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand.	3.1 Different repairs for spark advance system. 3.2 Identification and spark advance system. 3.3 Arithmetic operations 3.4 Mensuration 3.5 Use of measuring devices 3.6 Service/Repair Manual 3.7 Post repair testing for spark advance system. 3.8 OSHS 3.9 Wearing of PPEs 3.10 Accomplishment of Repair Order	3.1 Sourcing of information 3.2 Interpreting information from manufacturer and workshop literature. 3.3 Identifying faults and its cause 3.4 Applying safety practices 3.5 Applying arithmetic operations 3.6 Repairing spark advance system. 3.7 Performing post-repair testing 3.8 Accomplishing Repair Order

ELEMENT	PERFORMANCE CRITERIA ***Italicized terms*** are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	3.5 Documentation report is prepared based on analysis result. 3.6 Safety practices are applied following Occupational Health and Safety (OSH) procedure.		
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is turned-over to immediate superior for quality control following workplace procedure. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 ***Workplace documents*** are accomplished according to workplace procedures.	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements includes: 1.1 Inspect spark plug 1.2 Inspect and replace ignition coil 1.3 Inspect ignition timing 1.4 Inspect battery 1.5 Inspect timing belt 1.6 Inspect timing chain 1.7 Inspect sensors and actuators 1.8 Check, inspect and replace ECU
2. Industry criteria	Industry criteria may include: 2.1 Repair manual 2.2 Workplace procedures 2.3 Safety and environmental requirements 2.4 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Multi meter 3.1.4 Trouble light 3.2 Equipment: 3.2.1 Lifter 3.2.2 Scanner
4. Materials	Materials may include: 4.1 Rags 4.2 Grease 4.3 Penetrating oil 4.4 PPEs
5. Diagnostic procedures	Diagnostic procedures may include: 5.1 Manual diagnosis (symptoms and problems) 5.2 Diagnostic scan tool (diagnostic trouble code-DTC)
6. Repair for electronic spark advance system components	Repair for electronic spark advance system components may include: 6.1 Replace ignition coil 6.2 Replace spark plug 6.3 Replace related sensors
7. Functionality test	Functionality test may include: 7.1 Checking and analyzing data list 7.2 Check active/actuator test using diagnostic scan tool 7.3 Test drive
8. Workplace documents	Workplace documents may include: 8.1 Repair order 8.2 Inspection form

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair a spark advance system 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards associated with the work and managed risks. 1.1.4 Selected and checked tools and equipment. 1.1.5 Checked and reported availability of materials, parts and supplies. 1.1.6 Applied safety practices. 1.2 Diagnosed a spark advance system 1.2.1 Performed diagnostic procedures. 1.2.2 Analyzed diagnostic symptoms. 1.2.3 Reported inspection findings and recommendations. 1.2.4 Applied safety practices. 1.3 Repaired a spark advance system 1.3.1 Removed spark advance system component. 1.3.2 Carried out repair for electronic spark advance system components. 1.3.3 Carried out functionality test. 1.3.4 Analyzed and selected repair options and solutions. 1.3.5 Prepared documentation report. 1.3.6 Applied safety practices. 1.4 Completed work processes 1.4.1 Conducted final inspection. 1.4.2 Turned-over vehicle. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Accomplished workplace documents.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration with Oral questioning 3.2 Written exam 3.3 Direct Observation
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

SECTION 3 TRAINING ARRANGEMENTS

These guidelines are set to provide the Technical and Vocational Education and Training (TVET) providers with information and other important requirements to consider when designing training programs for **AUTOMOTIVE DIAGNOSIS (ENGINE) NC III**.

3.1 CURRICULUM DESIGN

TESDA shall provide the training on the development of competency-based curricula to enable training providers develop their own curricula with the components mentioned below.

Delivery of knowledge requirements for the basic, common and core units of competency specifically in the areas of mathematics, science/technology, communication/language and other academic subjects shall be contextualized. To this end, TVET providers shall develop a Contextual Learning Matrix (CLM) to accompany the curricula.

Course Title: **AUTOMOTIVE DIAGNOSIS (ENGINE)** **NC Level** **NC III**

Nominal Training Duration:

40	Hours (Basic Competencies)
162	Hours (Common Competencies)
300	Hours (Core Competencies)
502	**TOTAL HOURS**
150	SIL

Course Description:

This course is designed to enhance the knowledge, skills and attitudes of an individual in the field of automotive servicing in accordance with industry standards. It covers core competencies in diagnosing and repairing: petrol electronic fuel injection system, emission control system, diesel electronic fuel injection system, forced-induction system, and electronic spark advance system.

Upon completion of the course, the learners are expected to demonstrate the above- mentioned competencies to be employed.

To obtain this, all units prescribed for this qualification must be achieved.

BASIC COMPETENCIES
40 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Lead workplace communication	1.1 Communicate information about workplace processes	1.1.1 Read ○ Effective verbal communication methods ○ Sources of information	● Lecture	● Written Test	2 Hours
		1.1.2 Practice organizing information	● Demonstration	● Observation	
		1.1.3 Identify organization requirements for written and electronic communication methods	● Lecture	● Written Test	
		1.1.4 Follow organization requirements for the use of written and electronic communication methods	● Demonstration ● Practical exercises	● Observation	
		1.1.5 Perform exercises on understanding and conveying intended meaning scenario	● Demonstration ● Role Play	● Observation	
	1.2 Lead workplace discussions	1.2.1 Describe: ● Organizational policy on production, quality and safety ● Goals/ objectives and action plan setting	● Group discussion	● Oral evaluation	2 Hours
		1.2.2 Read ● Effective verbal communication methods	● Lecture	● Written Test	
		1.2.3 Prepare/set action plans based on organizational goals and objectives	● Demonstration	● Observation	

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.3 Identify and communicate issues arising in the workplace	1.2.4 Describe: ● Organizational policy in dealing with issues and problems	● Group discussion	● Oral evaluation	2 Hours
		1.2.5 Read ● Effective verbal communication methods	● Lecture	● Written Test	
2. Lead small teams	2.1 Provide team leadership	2.1.1 Discussion of Company policies and procedures 2.1.2 Read web pages on situational leadership 2.1.3Role play on situational leadership	● Group work ● Role Play ● Lecture/ Discussion ● Individual Work	● Role Play ● Written Test	1 hour
	2.2 Assign responsibilities	2.2.1 Read web pages on performance management 2.2.2 Case study on allocating roles and responsibilities based on competencies of current staff	● Individual Work ● Case Study	● Role Play ● Written Test	1 hour
	2.3 Set performance expectations for team members	2.3.1 Role play to communicate performance expectations with staff 2.3.2 Discussion on performance issues	● Lecture/ Discussion ● Role Play	● Role Play ● Written Test	1 hour
	2.4 Supervise team performance	2.4.1 Discussion on performance monitoring 2.4.2 Role play on providing feedback on performance 2.4.3 Role play on performance coaching 2.4.4 Discussion on keeping the team informed of team performance	● Lecture/ Discussion ● Role Play ● Case Study	● Role Play ● Written Test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		2.4.5 Case study on Team performance monitoring and feedback			
3. Apply critical thinking and problem-solving techniques in the workplace	3.1 Examine specific workplace strategies	3.1.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.1.2 Show mastery of the current industry hardware and software products and services 3.1.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.1.4 Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations ● Relevant equipment and operational processes ● Enterprise goals, targets and measures ● Enterprise quality OHS and environmental requirement ● Enterprise information systems and data collation ● Industry codes and standards	● Group discussion ● Lecture ● Demonstration ● Role playing	● Case Formulation ● Life Narrative Inquiry (Interview) ● Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.2 Analyze the causes of specific workplace challenges	3.2.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.2.2 Show mastery of the current industry hardware and software products and services 3.2.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.2.4 Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations ● Relevant equipment and operational processes ● Enterprise goals, targets and measures ● Enterprise quality OHS and environmental requirement ● Enterprise information systems and data collation ● Industry codes and standards 3.2.5 Identify extent and causes of specific challenges in the workplace 3.2.6 Use of range of analytical problem-solving techniques	● Group discussion ● Lecture ● Demonstration ● Role playing	● Case Formulation ● Life Narrative Inquiry (Interview) ● Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		3.2.7 Formulate clear-cut findings on the nature of each identified workplace challenges			
	3.3 Formulate resolutions to specific workplace challenges	3.3.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.3.2 Show mastery of the current industry hardware and software products and services 3.3.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.3.4.Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations ● Relevant equipment and operational processes ● Enterprise goals, targets and measures ● Enterprise quality OHS and environmental requirement ● Enterprise information systems and data collation ● Industry codes and standards 3.3.5 Identify extent and causes of specific challenges in the workplace	● Group discussion ● Lecture ● Demonstration ● Role playing	● Case Formulation ● Life Narrative Inquiry (Interview) ● Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		3.3.6 Use of range of analytical problem-solving techniques 3.3.7 Formulate clear-cut findings on the nature of each identified workplace challenges 3.3.8 Discus strategies on devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges			
	3.4 Implement action plans and communicate results	3.4.1 Identify extent and causes of specific challenges in the workplace 3.4.2 Use of range of analytical problem-solving techniques 3.4.3 Formulate clear-cut findings on the nature of each identified workplace challenges 3.4.4 Discus strategies on devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour
4. Work in a Diverse Environment	4.1 Develop an individual's cultural awareness and sensitivity	4.1.1 Show understanding of cultural diversity in the workplace 4.1.2 Recognize norms of behavior for interacting and dialogue with specific groups (e. g., Muslims and other non-Christians, non-Catholics, tribes/ethnic groups, foreigners)	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/ problems	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		4.1.3 Demonstrate different methods of verbal and non-verbal communication in a multicultural setting 4.1.4 Apply cross-cultural communication skills (i.e. different business customs, beliefs, communication strategies) 4.1.5 Show affective skills – establishing rapport and empathy, understanding, etc. 4.1.6 Demonstrate openness and flexibility in communication 4.1.7 Recognize diverse groups in the workplace and community as defined by divergent culture, religion, traditions and practices		involving workplace diversity issues • Written examination • Role Playing	
	4.2 Work effectively in an environment that acknowledges and values cultural diversity	4.2.1 Explain the value of diversity in the economy and society in terms of 4.2.2 Workforce development 4.2.3 Discuss the importance of inclusiveness in a diverse environment 4.2.4 Discuss the importance of shared vision and understanding of and commitment to team, departmental, and organizational goals and objectives	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/ problems involving workplace diversity issues • Written examination	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		4.2.5 Identify and exhibit strategies for customer service excellence 4.2.6 Demonstrate cross-cultural communication skills and active listening 4.2.7 Recognize diverse groups in the workplace and community as defined by divergent culture, religion, traditions and practices 4.2.8 Demonstrate collaboration skills		• Role Playing	
	4.3 Identify common issues in a multicultural and diverse environment	4.3.1 Explain the value, and leverage of cultural diversity 4.3.2 Discuss the inclusivity and conflict resolution 4.3.3 Describe the workplace harassment 4.3.4 Explain the change management and cite ways to overcome resistance to change 4.3.5 Demonstrate advanced strategies for customer service excellence 4.3.6 Address diversity-related conflicts in the workplace 4.3.7 Eliminate discriminatory behavior towards customers and co-workers 4.3.8 Utilize change management policies in the workplace	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/ problems involving workplace diversity issues • Written examination • Role Playing	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
5. Propose methods of applying learning and innovation in the organization	5.1 Assess work procedures, processes and systems in terms of innovative practices	5.1.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.1.2 Demonstrate collaboration and networking skills 5.1.3 Show basic skills in research 5.1.4 Generate practical insights on how to improve organizational procedures, processes and systems	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour
	5.2 Generate practical action plans for improving work procedures, processes	5.2.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.2.2 Demonstrate collaboration and networking skills 5.2.3 Show basic skills in research 5.2.4 Generate practical insights on how to improve organizational	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance.	1 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		procedures, processes and systems 5.2.5 Set up action plans on how to apply innovative procedures in the organization 5.2.6 Set up action plans on how to apply innovative procedures in the organization 5.2.7 Generate practical insights on how to improve organizational procedures, processes and systems		• Standardized assessment of character strengths and virtues applied	
	5.3 Evaluate the effectiveness of the proposed action plans	5.3.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.3.2 Demonstrate collaboration and networking skills 5.3.3 Show basic skills in research 5.3.4 Generate practical insights on continuous improvement	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
6. Use information systematically	6.1 Use technical information systems	6.1.1 Lecture and discussion on: ● Application in collating information ● Procedures for inputting, maintaining and archiving information ● Guidance to people who need to find and use information 6.1.2 Organizing information into a suitable form for reference and use 6.1.3 Classify stored information for identification and retrieval 6.1.4 Operate the technical information system by using agreed procedures	● Lecture ● Group Discussion ● Hands on ● Demonstration	● Oral evaluation ● Written Test ● Observation ● Presentation	4 Hours
	6.2 Apply information technology (IT) systems	6.2.1 Lecture and discussion on: ● Attributes and limitations of available software tool ● Procedures and work instructions for the use of IT ● Operational requirements for IT systems ● Sources and flow paths of data ● Security systems and measures that can be used ● Methods of entering and processing information 6.2.2. Use procedures and work instructions for the use of IT 6.2.3 Extract data and format reports	● Lecture ● Group Discussion ● Self-paced handout/ module ● Hands on ● Demonstration	● Oral evaluation ● Written Test ● Observation ● Presentation	2 Hours

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Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		6.2.4 Use WWW applications			
	6.3 Edit, format and check information	6.3.1 Lecture and discussion on: • Basic file-handling techniques • Techniques in checking documents • Techniques in editing and formatting • Proof reading techniques 6.3.2Use different techniques in checking documents 6.3.3 Edit and format information applying different techniques 6.3.4 Proof read information applying different techniques	• Lecture • Group Discussion • Self-paced handout/ module • Hands on • Demonstration	• Oral evaluation • Written Test • Observation • Presentation	2 Hours
7. Evaluate Occupational	7.1 Interpret Occupational	7.1.1 Discuss the OSH standards, principles and legislations	• Lecture • Group Discussion	• Written Exam • Demonstration	1.5 hr

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Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
Safety And Health Work Practices	Safety and Health practices	7.1.2 Identify OSH work practices issues 7.1.3 Discuss standard safety requirements		• Observation • Interviews / • Questioning	
	7.2 Set OSH work targets	7.2.1 Discussion in actions plans that are necessary in achieving the OSH target	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1 hr
	7.3 Evaluate effectiveness of Occupational Safety and Health work instructions	7.3.1 Practice evaluating safety data (Historical or Simulated)	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1.5 hr
8.Evaluate Environmental Work Practices	8.1 Interpret Environmental practices, policies and procedures	8.1.1 Discussion Environmental Issues regarding ● Water Quality ● National and Local Government Issues ● Safety ● Endangered Species ● Noise ● Air Quality ● Historic ● Waste ● Cultural 8.1.2 Updating of existing occupation practices	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hr
	8.2 Establish targets to evaluate	8.2.1 Discussion on ● lower production costs and energy consumption	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation	1 hr

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	environmental practices	● Environmentally Sound Processes ● Resource Efficient ● Recycling and Waste Management 8.2.2 Simple case study regarding energy efficiency		● Interviews / Questioning	
	8.3 Evaluate effectiveness of environmental practices	8.3.1 Identifying effective environmental practices relevant to the industry/occupation ● Implementation of energy efficiency	● Lecture ● Group Discussion ● Demonstration ● Case Study	● Written Exam ● Demonstration ● Observation ● Interviews / Questioning ● Third Party Reports	1 hr
9. Facilitate Entreprenuerial Skills For Micro-Small-Medium Enterprises (MSMEs)	9.1 Develop and maintain micro-small-medium enterprise (MSMEs) skills in the organization	9.1.1 Discussions on business models and strategies 9.1.2 Discussion on Types and categories of businesses and business internal control 9.1.3 Discussion on Relevant National and local legislations affecting businesses 9.1.4 Prepare promotional materials 9.1.5 Practice basic bookkeeping	● Lecture/ Discussion ● Case Study ● Demonstration	● Written Test ● Portfolio ● Work Related Project	2 hours
	9.2 Establish and Maintain client-base/market	9.2.1 Role play on customer and employee relations 9.2.2 Discussion on Basic product promotion strategies 9.2.3 Preparation of Basic Feasibility study 9.2.4 Case studies on Basic Business ethics	● Role Play ● Lecture Discussion ● Case study	● Case problem ● Written Test	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		9.2.5 Prepare basic advertising materials			
	9.3 Apply budgeting and financial management skills	9.3.1 Discussion on: ● Basic cost-benefit analysis ● Basic financial management ● Basic financial accounting ● Business internal controls	● Role Play ● Lecture Discussion ● Group work	● Written Test ● Case problem	1 hour

COMMON COMPETENCIES
162 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Validate vehicle specification	1.1 Check body type of the vehicle	● Enumerate the different kinds of vehicle ● Explain the difference of each kind of vehicle ● Identify the measuring points of the vehicle ● Explain the procedures in measuring vehicle dimension and weight ● Describe the different body shapes of the vehicle ● Differentiate kinds of power train ● Explain the function of each power train ● Discuss occupational safety and health standard in checking the body type of a vehicle ● Identify different kinds of vehicle ● Measure vehicle dimensions and weight ● Identify vehicle body shapes ● Identify vehicle power train ● Apply safety practices	● Lecture ● Demonstration ● Video presentation	● Written exam ● Demonstrate	7 hrs
	1.2 Check vehicle engine type	● Discuss the different kinds of engine ● Enumerate the different kinds of fuel/energy system ● Describe the different engine components ● Identify different kinds of engine	● Lecture ● Demonstration ● Video presentation	● Written exam ● Demonstrate	3 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		● Identify different types of fuel/energy system ● Identify different engine components			
	1.3 Check vehicle specifications	● Inspect VIN plate of the vehicle ● Verify vehicle specification ● Check vehicle modifications and conversions ● Inspect vehicle conversions ● Explain different vehicle related regulations in the Philippine	● Lecture ● Demonstration ● Video presentation	● Written exam ● Demonstrate	4 hrs
	1.4 Complete validation of vehicle specification	● Explain verification of vehicle ownership using repair order and vehicle reference materials ● Discuss procedures in accomplishing check sheet ● Discuss submission of check sheet	● Lecture ● Demonstration ● Video presentation	● Written exam ● Demonstrate	3 hrs
2. Move and position vehicle	2.1 Prepare vehicle for operation	● Explain vehicle multi point inspection ● Enumerate cockpit drill procedure ● Initialize engine startup ● Perform parking brake ● Show vehicle operational procedures	● Lecture discussion ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	16 hours
	2.2 Position vehicle	● Determine workshop hazards ● Discuss the procedure in avoiding workshop hazards ● Define occupational health and safety standards ● Move the vehicle ● Explain workshop rules and regulations	● Lecture ● Demonstration ● Video presentation	● Demonstration ● Written exam ● Interview	16 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.3 Park and stop the vehicle	● Explain parking rules and regulations ● Park vehicle ● Outline parking principles ● Shut-off vehicle	● Lecture ● Demonstration ● Video presentation	● Demonstration ● Written exam ● Interview	8 hours
3. Utilize automotive tools	3.1 Prepare automotive tools	● Identify and select automotive tools and attachments ● Discuss inspection and selection procedures ● Describe the defects and damages of automotive tools and attachments ● Discuss OSHS in preparation of automotive tools ● Prepare automotive tools and attachments	● Lecture ● Demonstration ● Visual aids ● Videos	● Written examination ● Interview ● Demonstration ● Practical examination	6 hrs
	3.2 Use automotive tools	● Discuss the procedure in mounting attachments to automotive tools ● Discuss the procedure in connecting the power supply to power tools ● Discuss the procedure in operating the power tools ● Discuss the utilization of hand tools ● Identify PPEs ● Discuss OSHS in using automotive tools ● Use automotive tools ● Use PPEs	● Lecture ● Demonstration ● Visual aids ● Videos	● Written examination ● Interview ● Demonstration ● Practical examination	6 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.3 Maintain automotive tools	● Discuss the procedure in cleaning, checking for serviceability, and storing of automotive tools and attachments ● Discuss the procedure in identifying and reporting defects and damages ● Discuss the proper waste segregation ● Demonstrate the proper maintenance of automotive tools ● Demonstrate disposal of wastes	● Lecture ● Visual aids ● Videos	● Written examination ● Demonstration	4 hrs
4. Perform mensuration and calculation	4.1 Select measuring instruments	● Describe measuring instruments ● Select measuring instruments ● Inspect and calibrate measuring instruments ● Report and return defective measuring instruments ● Demonstrate safety practices	● Demonstration ● Video presentation ● Lecture Discussion ● Workshop visit	● Demonstration ● Written exam ● Oral questioning	9 hours
	4.2 Carry out measurements and calculation	● Explain formulas for volume, areas, perimeters of plane and geometric figures ● Explain the procedure in reading tools' limit of accuracy ● Measure required automotive parts ● Read tools' limit of accuracy ● Inspect and calibrate measuring instruments	● Demonstration ● Video presentation ● Lecture Discussion ● Workshop visit	● Demonstration ● Written exam ● Oral questioning	29 hours
	4.3 Maintain measuring instruments	● Identify PPEs ● Discuss cleaning procedures of measuring instruments	● Demonstration ● Video presentation	● Demonstration ● Written exam ● Oral questioning	5 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		● Enumerate steps in storing instruments ● Wear PPEs ● Clean measuring instrument tools ● Re-inspect and re-calibrate measuring instruments	● Lecture Discussion		
5. Utilize workshop facilities and equipment	5.1 Perform pre-operation activities	● Identify different areas of an automotive service facilities ● Explain the preparation procedures of automotive service facilities ● Enumerate different equipment in the automotive service facilities ● Discuss the preparation procedures of equipment ● Describe minor repairs in automotive facilities and equipment ● Describe defective equipment ● Identify reporting procedures for defective equipment ● Discuss OSHS practices related to the preparation of facilities and equipment ● Prepare workshop facilities and equipment	● Lecture ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	9 hrs
	5.2 Use facilities and equipment	● Explain the operation of equipment according to operation manual ● Describe how facilities are utilized according to workshop procedures	● Lecture ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	5 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		• Explain how equipment performance is monitored following users' manual • Describe the monitoring of facilities functionalities following workplace procedures • Discuss how OSHS safety practices are applied			
	5.3 Conduct post-operation activities	• Explain how workshop facilities are restored according to good housekeeping • Discuss tools and equipment are cleaned and stored according to good housekeeping • Explain wastes disposed following waste management procedure and OSHS • Enumerate the safety practices that are applied following OSHS • Demonstrate preparation of report based on workshop standard procedure	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	5 hours
6. Prepare servicing parts and consumables	6.1 Identify parts and consumables	• Familiarize parts & consumables • Identify indirect materials • Identify hazardous parts and consumables	• Lecture • Video presentation • Actual training	• Demonstration • Written exam • Interview	6 hrs
	6.2 Retrieve and withdraw parts and consumables	• Familiarize requisition slip • Perform parts withdrawal procedure & recording • Validate parts and consumables according to quantity & specification • Perform safety precautions	• Lecture • Video presentation • Actual training	• Demonstration • Written exam • Interview	4 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	6.3 Complete work process	• Segregate parts to be returned to customers • Segregate parts & consumables for proper disposal or recycling according to 3Rs and RA 6969 • Wear PPE's	• Lecture • Video presentation • Actual training	• Demonstration • Written exam • Interview	3 hrs
7. Prepare vehicle for servicing and releasing	7.1 Receive vehicle	• Identify different areas of an automotive service facility • Explain the receiving procedures of automotive service facilities • Explain the checklisting procedures of automotive service facilities • Describe minor repairs in automotive facilities and equipment • Discuss OSHS practices related to the preparation of facilities and equipment • Prepare workshop facilities and equipment	• Lecture • Demonstration • Video presentation • Workshop visit	• Role-playing • Written exam • Interview	6 hours
	7.2 Prepare vehicle for servicing	• Prepare vehicle for servicing • Explain the preparation procedures of automotive service facilities • Demonstrate the procedure in installing protective covers • Explain the concept of the locator blocks • Classify the type of vehicle repair based on the Repair Order	• Lecture • Demonstration	• Role-playing • Written Exams • Oral Exams	5 hours
	7.3 Prepare vehicle for releasing	• Use the repair order to identify work performed	• Lecture • Demonstration	• Role-Playing • Written Exams	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		● Apply quality control measures on work done ● Operate vehicle for transfer and release		● Oral Exams	

CORE COMPETENCIES
300 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Diagnose and repair petrol electronic fuel injection system	1.1 Prepare to diagnose electronic fuel injection system	1.1.1 Discuss and explain the following: ● OSHS ● Trouble shooting guide. ● Waste management ● Sourcing out and interpretation of repair information ● Service/Repair manual ● Tools, equipment, parts and supplies in inspecting and repairing electronic fuel injection system. ● Different job requirements ● Serviceability of tools and equipment ● Work hazards ● Repair Order checklist ● Identification and function of electronic fuel injection system. 1.1.2 Prepare to diagnose electronic fuel injection system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	**TOTAL = 80 Hours** 8 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.2 Perform diagnosis	1.2.1 Discuss and explain the following: ● Interpretation of job requirements ● Identification and function of electronic fuel injection system. ● Trouble shooting guide ● Related system analysis ● Mensuration ● Arithmetic operations ● Use of measuring devices ● Operate diagnostic tools ● Preparation of report on inspection findings, recommendation and repair instructions ● Diagnostic procedures ● OSHS ● Wearing of PPEs ● Industry criteria 1.2.2 Perform diagnosis	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	32 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.3 Repair electronic fuel injection system	1.3.1 Discuss and explain the following: ● Different repairs for electronic fuel injection system ● Identification and function of electronic fuel injection system. ● Arithmetic operations ● Mensuration ● Use of measuring devices ● Service/Repair Manual ● Functionality test for electronic fuel injection system. ● Isolation and elimination approach ● Waste management ● OSHS ● Wearing of PPEs ● Accomplishment of Repair Order 1.3.2 Repair electronic fuel injection system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	32 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.4 Complete work processes	1.4.1 Discuss and explain the following: ● Final inspection procedure: ● Visual inspection ● Checking of tightening of torque ● Turn-over of vehicle ● Accomplishment of repair order and other forms ● Job done ● OSHS ● Wearing of PPEs ● Waste Management ● 3Rs ● 5S ● Checking and storage of tools and equipment ● Workplace documents 1.4.2 Complete work processes	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	8 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
2. Diagnose and repair emission control system	2.1 Prepare to diagnose and repair vehicle emission control system	2.1.1 Discuss and explain the following: ● OSHS ● Trouble shooting guide. ● Waste management ● Sourcing out and interpretation of repair information ● Service/Repair manual ● Tools, equipment materials, parts and supplies in inspecting and repairing faults on emission control system ● Different job requirements ● Serviceability of tools and equipment ● Work hazards ● Repair Order checklist ● Identification and function of emission control system. 2.1.2 Prepare to diagnose and repair vehicle emission control system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	**TOTAL = 40 Hours** 4 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.2 Perform diagnosis	2.2.1 Discuss and explain the following: ● Interpretation of job requirements ● Identification and function of emission control system. ● Trouble shooting guide ● Related system analysis ● Mensuration ● Arithmetic operations ● Diagnostic procedures ● Use of measuring devices ● Operate diagnostic tools ● Reporting procedures ● Preparation of report on inspection findings, recommendation and repair instructions ● Awareness on the use of opacity meter ● OSHS ● Wearing of PPEs ● Industry criteria 2.2.2 Perform diagnosis	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	16 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.3 Repair emission control system	2.3.1 Discuss and explain the following: ● Different repairs for emission control system ● Identification and function of emission control system ● Arithmetic operations ● Mensuration ● Use of measuring devices ● Service/Repair Manual ● Functionality test for electronic fuel injection system. ● OSHS ● Wearing of PPEs ● Accomplishment of Repair Order ● Isolation and elimination approach ● Waste management 2.3.2 Repair emission control system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	16 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.4 Complete work processes	2.4.1 Discuss and explain the following: ● Final inspection procedure: ● Visual inspection ● Checking of tightening of torque ● Turn-over of vehicle ● Accomplishment of repair order and other forms ● Job done ● OSHS ● Wearing of PPEs ● 5S ● Waste management ● 3Rs ● Checking and storage of tools and equipment ● Workplace documents 2.4.2 Complete work processes	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	4 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
3. Diagnose and repair diesel electronic fuel injection system	3.1 Prepare to diagnose and repair diesel fuel injection system and common-rail EFI fuel injection system	3.1.1 Discuss and explain the following: ● OSHS ● Trouble shooting guide. ● Waste management ● Sourcing out and interpretation of repair information ● Service/Repair manual ● Tools, equipment materials, parts and supplies in inspecting and repairing faults on fuel injection system and common-rail EFI fuel injection system. ● Different job requirements ● Serviceability of tools and equipment ● Work hazards ● Repair Order checklist ● Identification and function of diesel fuel injection system and common-rail EFI fuel injection system. 3.1.2 Prepare to diagnose and repair diesel fuel injection system and common-rail EFI fuel injection system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	**TOTAL = 80 Hours** 8 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.2 Perform diagnosis	3.2.1 Discuss and explain the following: ● Interpretation of job requirements ● Diagnostic procedures ● Diagnose and repair common-rail EFI fuel injection system ● Identification and function of fuel injection system and common-rail EFI fuel injection system ● Trouble shooting guide ● Related system analysis ● Mensuration ● Arithmetic operations ● Use of measuring devices ● Operate diagnostic tools ● Preparation of report on inspection findings, recommendation and repair instructions ● OSHS ● Wearing of PPEs ● Industry criteria 3.2.2 Perform diagnosis	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	32 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.3 Repair diesel fuel injection system and common-rail EFI fuel injection system	3.3.1 Discuss and explain the following: ● Different repairs for fuel injection system and common-rail EFI fuel injection system. ● Identification and function of fuel injection system and common-rail EFI fuel injection system ● Arithmetic operations ● Mensuration ● Use of measuring devices ● Service/Repair Manual ● Functionality test for fuel injection system and EFI fuel injection system. ● Isolation and elimination approach ● Waste management ● OSHS ● Wearing of PPEs ● Accomplishment of Repair Order 3.3.2 Repair diesel fuel injection system and common-rail EFI fuel injection system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	32 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.4 Complete work processes	3.4.1 Discuss and explain the following: ● Final inspection procedure: - Visual inspection - Checking of tightening of torque ● Turn-over of vehicle ● Accomplishment of repair order and other forms - Job done ● OSHS ● Wearing of PPEs ● 5S ● Waste management ● 3Rs ● Checking and storage of tools and equipment ● Workplace documents 3.4.2 Complete work processes	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	8 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
4. Diagnose and repair forced-induction system	4.1 Prepare to diagnose and repair a forced-induction system	4.1.1 Discuss and explain the following: ● OSHS ● Trouble shooting guide. ● Diagnose and repair forced induction system. ● Waste management ● Sourcing out and interpretation of repair information ● Service/Repair manual ● Tools, equipment and materials in inspecting and repairing faults on forced induction system. ● Interpretation of job requirements ● Different job requirements ● Serviceability of tools and equipment ● Work hazards ● Accomplishment of Repair Order checklist ● Identification and function forced- induction system 4.1.2 Prepare to diagnose and repair a forced- induction system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	**TOTAL = 60 Hours** 6 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	4.2 Diagnose a forced-induction system	4.2.1 Discuss and explain the following: ● Identification and function forced induction system. ● Diagnostic procedures ● Trouble shooting guide ● Mensuration ● Arithmetic operations ● Use of measuring devices ● Reporting procedures ● OSHS ● Wearing of PPEs ● Industry criteria 4.2.2 Diagnose a forced- induction system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	24 Hours
	4.3 Repair forced-induction system	4.3.1 Discuss and explain the following: ● Different repairs for forced induction system. ● Identification and forced induction system. ● Arithmetic operations ● Mensuration ● Use of measuring devices ● Service/Repair Manual ● Post repair testing for fuel injection system. ● OSHS ● Wearing of PPEs ● Accomplishment of Repair Order 4.3.2 Repair forced-induction system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	24 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	4.4 Complete work processes	4.4.1 Discuss and explain the following: ● Final inspection procedure: - Visual inspection - Checking of tightening of torque ● Turn-over of vehicle ● Accomplishment of repair order and other forms - Job done ● OSHS ● Wearing of PPEs ● 3Rs ● 5S ● Waste management ● Checking and storage of tools and equipment ● Workplace documents 4.4.2 Complete work processes	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	6 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
5. Diagnose and repair electronic spark advance system	5.1 Prepare to diagnose and repair a spark advance system	5.1.1 Discuss and explain the following: ● OSHS ● Trouble shooting guide. ● Diagnose and repair spark advance system. ● Waste management ● Sourcing out and interpretation of repair information ● Service/Repair manual ● Tools, equipment and materials in inspecting and repairing faults on spark advance system. ● Interpretation of job requirements ● Different job requirements ● Serviceability of tools and equipment ● Work hazards ● Accomplishment of Repair Order checklist ● Identification sparks advance system 5.1.2 Prepare to diagnose and repair a spark advance system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	**TOTAL = 40 Hours** 4 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.2 Diagnose a spark advance system	5.2.1 Discuss and explain the following: ● Identification and function spark advance system. ● Diagnostic procedures ● Trouble shooting guide ● Mensuration ● Arithmetic operations ● Use of measuring devices ● Use of scan tools ● Reporting procedures ● OSHS ● Wearing of PPEs ● Industry criteria 5.2.2 Diagnose a spark advance system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	16 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.3 Repair a spark advance system	5.3.1 Discuss and explain the following: ● Different repairs for spark advance system. ● Identification and spark advance system. ● Arithmetic operations ● Mensuration ● Use of measuring devices ● Service/Repair Manual ● Post repair testing for spark advance system. ● OSHS ● Wearing of PPEs ● Accomplishment of Repair Order 5.3.2 Repair a spark advance system	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	16 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.4 Complete work processes	5.4.1 Discuss and explain the following: ● Final inspection procedure: - Visual inspection - Checking of tightening of torque ● Turn-over of vehicle ● Accomplishment of repair order and other forms - Job done ● OSHS ● Wearing of PPEs ● 3Rs ● 5S ● Waste management ● Checking and storage of tools and equipment ● Workplace documents 5.4.2 Complete work processes	● Lecture ● Video presentation ● Demonstration	● Demonstration ● Written exam ● Interview	4 Hours

3.2 TRAINING DELIVERY

1. **The delivery of training shall adhere to the design of the curriculum. Delivery shall be guided by the principles of competency-based TVET.**

   a. Course design is based on competency standards set by the industry or recognized industry sector; (Learning system is driven by competencies written to industry standards)
   b. Training delivery is learner-centered and should accommodate individualized and self-paced learning strategies;
   c. Training can be done on an actual workplace setting, simulation of a workplace and/or through adoption of modern technology.
   d. Assessment is based in the collection of evidence of the performance of work to the industry required standards;
   e. Assessment of competency takes the trainee's knowledge and attitude into account but requires evidence of actual performance of the competency as the primary source of evidence.
   f. Training program allows for recognition of prior learning (RPL) or current competencies;
   g. Training completion is based on satisfactory performance of all specified competencies.

2. **The competency-based TVET system recognizes various types of delivery modes, both on-and off-the-job as long as the learning is driven by the competency standards specified by the industry. The following training modalities and their variations/components may be adopted singly or in combination with other modalities when designing and delivering training programs:**

   2.1 School/Institution- Based:

   - Dual Training System (DTS)/Dualized Training Program (DTP) which contain both in-school and in-industry training or fieldwork components. Details can be referred to the Implementing Rules and Regulations of the DTS Law and the TESDA Guidelines on the DTP.

   - Distance learning is a formal education process in which majority of the instruction occurs when the students and instructor are not in the same place. Distance learning may employ correspondence study, audio, video, computer technologies or other modern technologies that can be used to facilitate learning and formal and non-formal training. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

   - Supervised Industry Training (SIT) or on-the-job training (OJT) is an approach in training designed to enhance the knowledge and skills of the trainee through actual experience in the workplace to acquire specific competencies as prescribed in the training regulations. It is imperative that the deployment of trainees in the workplace is adhered to training programs agreed by the institution and enterprise and status and progress of trainees are closely monitored by the training institutions to prevent opportunity for work exploitation.

- The classroom-based or in-center instruction uses of learner-centered methods as well as laboratory or field-work components.

2.2 Enterprise-Based:

- **Formal Apprenticeship** – Training within employment involving a contract between an apprentice and an enterprise on an approved apprenticeable occupation.

- **Informal Apprenticeship** - is based on a training (and working) agreement between an apprentice and a master craftsperson wherein the agreement may be written or oral and the master craftsperson commits to training the apprentice in all the skills relevant to his or her trade over a significant period of time, usually between one and four years, while the apprentice commits to contributing productively to the work of the business. Training is integrated into the production process and apprentices learn by working alongside the experienced craftsperson.

- **Enterprise-based Training** - where training is implemented within the company in accordance with the requirements of the specific company. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

2.3 Community-Based – short term program conducted by non-government organizations (NGOs), LGUs, training centers and other TVET providers which are intended to address the specific needs of a community. Such programs can be conducted in informal settings such as barangay hall, basketball courts, etc. These programs can also be mobile training program (MTP)

3.3 TRAINEE ENTRY REQUIREMENTS

Trainees or students who would like to enroll in this program must possess the following requirements:

- Must be holder of Automotive Servicing (Engine Repair) NC II;
- Can communicate both oral and written;
- Can perform basic mathematical computation;
- Must be able to drive a vehicle; and
- Must possess a valid driver license.

This list does not include specific institutional requirements such as educational attainment, appropriate work experience, and others that may be required of the trainees by the school or training center delivering the TVET program.

3.4 TOOLS, EQUIPMENT AND MATERIALS

AUTOMOTIVE DIAGNOSIS (ENGINE) NC III

Recommended list of tools, equipment and materials for the training of 25 trainees for Automotive Diagnosis (Engine) NC III.

Up-to-date tools, materials, and equipment of equivalent functions can be used as alternatives.

TOOLS	
QTY	**DESCRIPTION**
5 sets	Standard technician hand tools
1 set	Fuel pressure gauge set includes: - adaptor - hose - gauge - accessories - coupler
3 units	Multi-tester (analog)
3 units	Multi-tester (digital), 12V
2 pcs	Trouble light, medium size
5 sets	Soldering kit set includes: - desoldering pump - soldering lead - soldering paste - soldering iron holder - soldering iron
1 pc	Graduated cylinder, glass, 50ml
1 pc	Graduated cylinder, clear plastic, 50ml
1 pc	Vacuum tester
1 unit	Dial gauge with magnetic stand

TOOLS	
QTY	**DESCRIPTION**
25 pcs	Test harness (SST)
25 pcs	Back probe, assorted sizes (SST)
1 pc	Torque wrench (30 kgf- cm)
1 pc	Torque wrench (230 kgf- cm)
1 pc	Torque wrench (460 kgf- cm)
1 pc	Torque wrench (920 kgf- cm)
1 pc	Torque wrench (1900 kgf- cm)
1 pc	Torque wrench (2800 kgf- cm)
1 pc	Torque wrench (30 kgf- cm)
5 pcs	Spark plug wrench
1 unit	Fuel injector tester
4 pcs	Pin punch (5mm)*
4 pcs	Plastic hammer
5 pcs	Ball peen hammer, 1lb
4 pcs	Micrometer (0 – 25mm)
4 pcs	Caliper gauge
4 pcs	Inside diameter measuring tools
1 set	Drill bit set
1 set	Screw extractor set
2 pcs	Steering wheel puller
2 pcs	Torx driver (6mm)
2 pcs	Torx driver (8mm)
4 pcs	Rack & pinion holder
4 pcs	Steering adjustable wrench
4 pcs	Hexagon wrench (24mm)
4 pcs	Cylinder stopper wrench
4 pcs	Pinion shaft wrench
4 pcs	Stainless Steel Ruler, 12”
4 pcs	Union nut wrench (10mm)
4 pcs	Coil spring compressor
4 pcs	Ball joint puller
1 set	Clutch aligner
1 pc	Grease gun
2 pcs	Clutch guide
1 pc	Tire gauge
5 pcs	Trouble light
5 pcs	Feeler gauge
2 pcs	C-clamp
2 pcs	Wheel wedge
1 pc	Pilot bearing puller
4 pcs	Brake bleeder
1 pc	Tire pressure gauge, ball pen type
2 pcs	Flywheel pilot bearing puller
2 pcs	SST (prevent oil leakage from transmission)*

EQUIPMENT	
QTY	**DESCRIPTION**
1 unit	Training vehicle, petrol, with EFI
1 unit	Training vehicle, diesel, with common-rail
1 unit	Mock-up training engine for conventional diesel EFI
1 unit	Lifter, 2-post
1 unit	Diagnostic scan tool with oscilloscope
1 unit	Air Compressor

MATERIALS	
QTY	**DESCRIPTION**
5 K	Rags
2 sets	Vehicle protective kit set includes: - seat cover - steering wheel cover - fender cover - shifting knob/ lever cover - paper matting
5 rolls	Electrical tape, large
1 set	Rubber O-ring
5 cans	Contact cleaners, 350ml/can
10 feet	Shrinkable tube, wire size #14
10 feet	Shrinkable tube, wire size #16
2 packs	Cable tie, 8”
2 sets	Cable tie, assorted sizes
1 can	Grease, 350ml/can
1 can	Penetrating oil, 350ml/can
4 L	Kerosene
10 pcs	Sand paper, 1,000 grit
1 pc	Gasket
	PPEs:
25 pcs	Apron
25 pairs	Gloves

NOTE: Access to and use of equipment/facilities can be provided through cooperative arrangements or MOA with other partner/companies.

3.5 TRAINING FACILITIES

AUTOMOTIVE DIAGNOSIS (ENGINE) NC III

Based on a class intake of 25 learners/trainees.

SPACE REQUIREMENT	SIZE IN METERS	AREA IN SQ. METERS	TOTAL AREA IN SQ. METERS
A. Building (permanent)			
● Lecture Room	5 x 6	30.00	30.00
● Workshop/ Laboratory area (Work bay)	4 x 7	28 x 2	56.00
● Storage Area (Tool room & S/M storage area)		20	20.00
● Learning Resource Center	5 x 4	20.00	20.00
● Wash area/ comfort room (Male, Female, PWD)		12	12.00
TOTAL			**138.00**

Note: Access to and use of equipment and facilities can be provided through cooperative arrangements of MOA with other partner enterprises/organizations/institutions.

3.6 TRAINERS' QUALIFICATIONS

NEW TRAINER

- Holder of National TVET Trainers Certificate (NTTC) Level 1 in Automotive Diagnosis (Engine) NC III or Automotive Servicing NC IV; and
- Must have at least 1-year industry experience on Automotive Diagnosis within the last 3 years.

EXISTING TRAINER

- Holder of National TVET Trainers Certificate (NTTC) Level 1 in Automotive Diagnosis (Engine) NC III or Automotive Servicing NC IV; and
- Must have industry immersion of 80 hours annually (industry training which includes structured training program inclusive of hands-on activities, observation in a workshop, and training certificates with number of hours).

3.6 INSTITUTIONAL ASSESSMENT

Institutional Assessment is gathering of evidence to determine the achievements of the requirements of the qualification to enable the trainer to make judgment whether the trainee is competent or not yet competent.

SECTION 4 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

Competency Assessment is the process of collecting evidence and making judgments whether competency has been achieved. The purpose of assessment is to confirm that an individual can perform to the standards expected at the workplace as expressed in relevant competency standards.

The assessment process is based on evidence or information gathered to prove achievement of competencies. The process may be applied to a full qualification or employable unit(s) of competency in partial fulfillment of the requirements of the national qualification.

4.1 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

4.1.1 To attain the national qualification of AUTOMOTIVE DIAGNOSIS (ENGINE) NC III, the candidate must demonstrate competence in all units listed in Section 1. Successful candidates shall be awarded a National Certificate signed by the TESDA Director General.

4.1.2 Assessment shall cover all competencies with basic and common integrated or assessed concurrently with the core units of competency.

4.1.3 The following are qualified to apply for assessment and certification, as long as they are holders of National Certificate in the amended Automotive Servicing (Engine Repair) NC II:

4.1.3.1 Graduates of WTR-registered program on Automotive Diagnosis (Engine) NC III, or graduates of NTR programs or formal/non-formal/informal including enterprise-based training programs related to automotive servicing (engine repair); or

4.1.3.2 Candidates who gained competencies in implementing automotive engine diagnosis on vehicles through informal training or previous work experiences. For informal training, a certification from any government agency or LGU or Barangay indicating the competencies of an individual. For work experience, a certificate of Employment and Job Description must be provided as proof.

4.1.4 Current holders of National Certificate (NC) and Certificate of Competency (COC) in AUTOMOTIVE SERVICING NC III shall have to undergo reassessment in the amended Training Regulations upon expiration of their Certificates. Candidates must be a holder of a National Certificate in the amended Automotive Servicing (Engine Repair) NC II.

4.1.5 Recognition of Prior Learning (RPL). Candidates who have gained competencies through informal training, previous work or life experiences may apply for recognition in a particular qualification through competency based on the current TESDA circular and/or implementing guidelines on portfolio assessment and RPL.

4.1.6 The industry shall determine assessment and certification requirements for each qualification with promulgated Training Regulations. It includes the following:

a. entry requirements for candidates
b. evidence gathering methods
c. qualification requirements of competency assessors
d. specific assessment and certification arrangements as by industry.

4.2 Competency Assessment Requisite

4.2.1 Self-Assessment Guide. The self-assessment guide (SAG) is accomplished by the candidate prior to actual competency assessment. SAG is a pre-assessment tool to help the candidate and the assessor determine what evidence is available, where gaps exist, including readiness for assessment.

This document can:

a. Identify the candidate’s skills and knowledge
b. Highlight gaps in candidate’s skills and knowledge
c. Provide critical guidance to the assessor and candidate on the evidence that need to be presented
d. Assist the candidate to identify key areas in which practice is needed or additional information or skills that should be gained prior

4.2.2 Accredited Assessment Center. Only Assessment Center accredited by TESDA is authorized to conduct competency assessment. Assessment centers undergo a quality assured procedure for accreditation before they are authorized by TESDA to manage the assessment for National Certification.

4.2.3 Accredited Competency Assessor. Only accredited competency assessor is authorized to conduct assessment of competence. Competency assessors undergo a quality assured system of accreditation procedure before they are authorized by TESDA to assess the competencies of candidates for National Certification.

COMPETENCY MAP
AUTOMOTIVE DIAGNOSIS (ENGINE) NC III

BASIC COMPETENCIES

Receive and respond to workplace communication	Participate in workplace communication	Lead workplace communication	Utilize specialized communication skill	Manage and sustain effective communication strategies
Work with others	Work in a team environment	Lead small teams	Develop and lead teams	Manage and sustain high performing teams
Solve/address routine problems	Solve/address general workplace problems	Apply critical thinking and problem-solving techniques in the workplace	Perform higher-order thinking processes and apply techniques in the workplace	Evaluate higher order thinking skills and adjust problem solving techniques
Enhance self-management skills	Develop career and life decisions	Work in a diverse environment	Contribute to the practice of social justice in the workplace	Advocate strategic thinking for global citizenship
Support innovation	Contribute to workplace innovation	Propose methods of applying learning and innovation in the organization	Manage innovative work instructions	Incorporate innovation into work procedures
Access and maintain information	Present relevant information	Use information systematically	Manage and evaluate usage of information	Develop systems in managing, and maintaining information
Follow occupational safety and health policies and procedures	Practice occupational safety and health policies and procedures	Evaluate occupational safety and health work practices	Lead in improvement of occupational safety and health program, policies and procedures	Manage implementation of OSH programs in the workplace
Apply environmental work standards	Exercise efficient and effective sustainable practices in the workplace	Evaluate environmental work practices	Lead towards improvement of environmental work programs, policies and procedures	Manage implementation of environmental programs in the workplace
Adopt entrepreneurial mindset in the workplace	Practice entrepreneurial skills in the workplace	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)	Sustain entrepreneurial skills	Develop and sustain a high-performing enterprise

COMMON COMPETENCIES

Validate vehicle specification	Move and position vehicle	Utilize automotive tools	Perform mensuration and calculation	Utilize workshop facilities and equipment
Prepare servicing parts and consumables	Prepare vehicle for servicing and releasing			

CORE COMPETENCIES

Diagnose and repair petrol electronic fuel injection system	Diagnose and repair emission control system	Diagnose and repair diesel electronic fuel injection system	Diagnose and repair forced-induction system	Diagnose and repair electronic spark advance system

GLOSSARY OF TERMS

1) Adjustment	a small alteration or movement made to achieve a desired fit, appearance, or result.
2) Bleeding	Refers to removal of air in fuel line from tank to filter.
3) Catalytic converter	refers to the process of either oxidation (diesel), deoxidation (petrol), or three-way catalytic (petrol).
4) Defective	Refers to problems to software and programs such as corruption.
5) Diagnose	identify the nature of problem by inspection of the symptoms.
6) Diagnostic symptoms	a physical manifestation that is regarded as indicating a condition of malfunction.
7) Evaluation of components	the making of a judgment about the condition of a part/component.
8) Final inspection	Includes road testing, oil leakage, functionality, etc.
9) Maintenance	the regular or periodic maintenance servicing of vehicles to keep it in top condition.
10) Out of standard	worn-out, unserviceable components, not conforming to manufacturer's standard.
11) Overhaul	take apart a major automobile component in order to examine it and repair/replace a part if necessary to bring back the major component to working conditions.
12) Petrol	also known as gasoline.
13) Repair	to apply any remedy to solve the problem.(tightening, cleaning, replacing, removal, installation, checking)
14) Service	the act of rendering maintenance service and repair/replacement of parts of an automobile to keep it in top condition.

TRAINING REGULATIONS (TR)
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THE TECHNICAL EXPERT PANEL (TEP)

MR. ROMMEL O. CABANELA Technical Expert Suzuki Philippines Inc. (SPH) Sub Com Chair for Training & Education Chamber of Automotive Manufactures of the Phils. Inc. (CAMPI)	**MR. GIDEON C. BRUNO** Mitsubishi Motors Philippines Corp. Member- Aftersales Committee Chamber of Automotive Manufactures of the Phils. Inc. (CAMPI)
MR. ELMER DEL ROSARIO Toyota Motor Philippines Corp. Member- Aftersales Committee Chamber of Automotive Manufactures of the Phils. Inc. (CAMPI)	**MR. GERARDO E. DAVID** Technical Expert Universal Motor Corp. Member- Aftersales Committee Chamber of Automotive Manufactures of the Phils. Inc. (CAMPI)

THE PARTICIPANTS IN THE NATIONAL VALIDATION OF THIS TRAINING REGULATION

Validators	Company
1. Mr. Lixper James B. Beleno	Isabela School of Arts and Trades
2. Mr. Rodrigo C. Montemayor	Gonzalo Puyat School of Arts and Trades
3. Mr. Rolando C. Escobar	Toyota Motor Philippines School of Technology, Sta. Rosa, Laguna
4. Mr. Fernan R. Rotarta	Toyota Motor Philippines School of Technology, Sta. Rosa, Laguna
5. Mr. Jovert M. Cabaces	RTC-CALABARZON
6. Mr. Francisco J. Reyes	TESDA Laguna
7. Mr. Felipe S. Gado	Romblon National Institute of Technology
8. Mr. Reynaldo M. Hernandez	Simeon Suan Vocational Technical College
9. Mr. Joefel N. Flestado	RTC-Pili
10. Mr. Ernie B. Pedroso	Passi Trade School
11. Mr. Dante L. Lagos	Leon Ganzon Polytechnic College
12. Engr. Emmanuel M. Garcia	Regional Training Center-Leyte
13. Engr. Augustine H. Matbagon	PTC-Valencia
14. Mr. Maynard C. Jumawan	PTC-Plaridel

Validators	Company
15. Mr. Alex P. Sabella	RTC-Iligan
16. Mr. Armando D. Bolivar	RTC-KPVTC
17. Mr. Raymund Luczon	Carmelo delos Cientos National Agricultural School
18. Mr. Henry Causing	PTC-Cotabato
19. Mr. Ricardo D. Bacaling	Gen. Santos National School of Arts and Trades
20. Mr. Jerry G. Lazatin	Honda Cars, Quezon City
21. Ms. Wea San Juan	Honda Cars, Manila
22. Mr. Henry Aguila	Honda Cars Kalookan Inc.
23. Mr. Jeffrey Resuello	Toyota, Bonifacio Global City, Taguig City
24. Mr. Dexter A. Reginio	Isuzu Philippines Corporation, Biñan, Laguna
25. Mr. Gino V. Batan	Isuzu Philippines Corporation, Alabang
26. Mr. John Paul M. Dador	Gateway, Pasay
27. Mr. Amado T. Antones, Jr.	Gateway, Pasay
28. Mr. Kurt Marcel A. Peñaflor	Gateway, Pasay
29. Mr. Jorlan B. Peligao	Gateway, Pasay
30. Mr. Mike Jayson T. Clarion, Iv	Gateway, Pasay
31. Mr. Mathew Emmanuel B. Lucena	Nissan, Otis
32. Mr. Argel Joseph S. Salvador	Nissan
33. Mr. Wilbert O. Tolosa	Diamond Motor Corporation, Cainta, Rizal
34. Mr. Vicente L. Geron	Nissan, Shaw

The Members of the TESDA Board and Secretariat

The MANAGEMENT and STAFF of the TESDA Secretariat

- Qualification and Standards Office
- Competency Standards Development Division
  - **MS. BERNADETTE S. AUDIJE**
  - **MS. CHERRY L. TORALDE**
  - **MS. MELCHRIS A. ATIS**

- Competency Programs and Standards Development Division
  - **MS. MERCEDES E. JAVIER**
  - **MR. VENZEL Y. CONCOLES**

Training Regulation are electronic copies.
For more information, please contact:

Technical Education and Skills Development Authority (TESDA)
Telefax No.: 8818-7728
or visit our website: www.tesda.gov.ph